Project Management

Syed Mahbubur Rahman

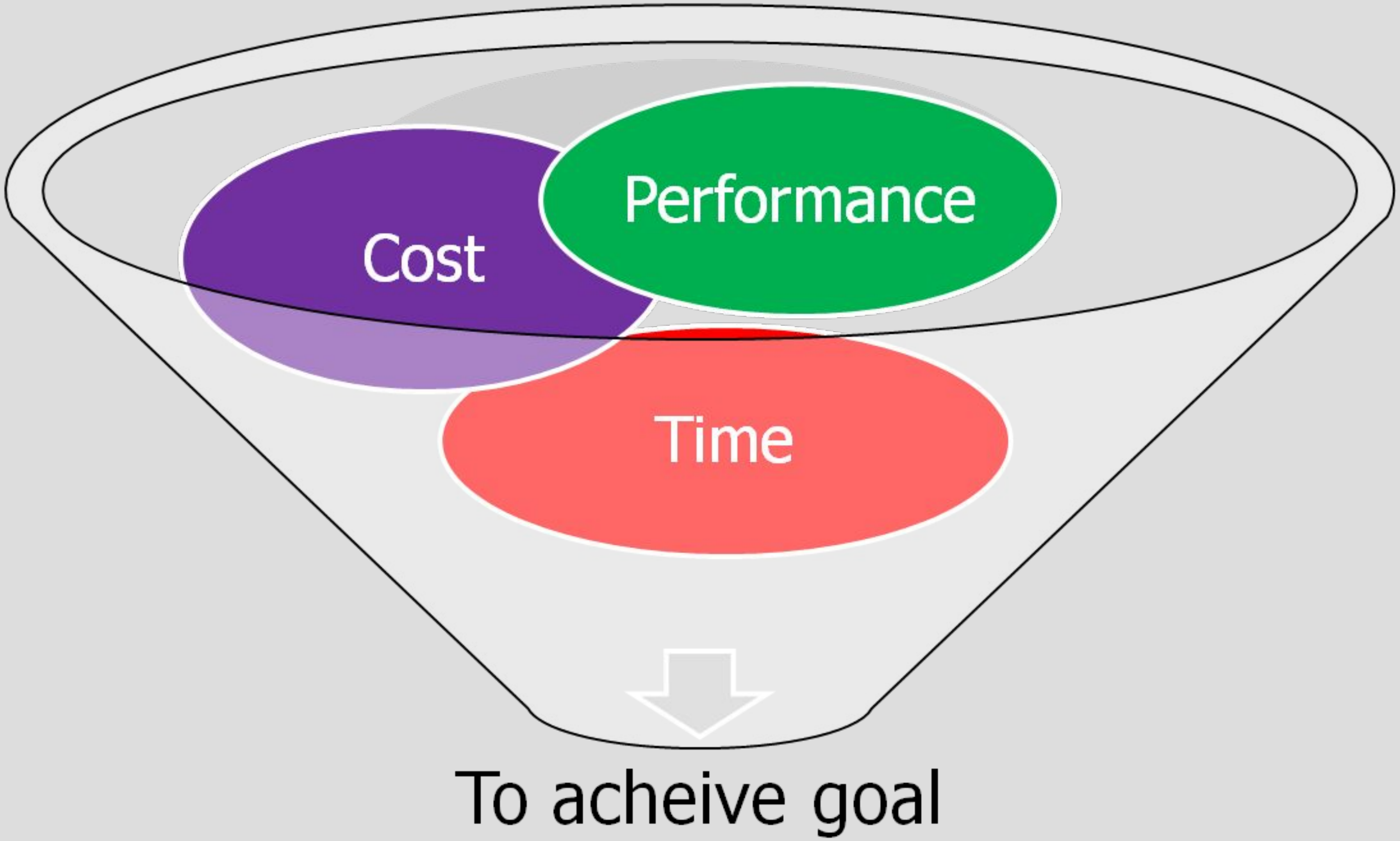
What is Project?

- A temporary endeavor undertaken to create a unique product or service

(PMI, 2001)

A set of activities with an aim to acheive a certain goal within stipulated time frame consuming specific resources

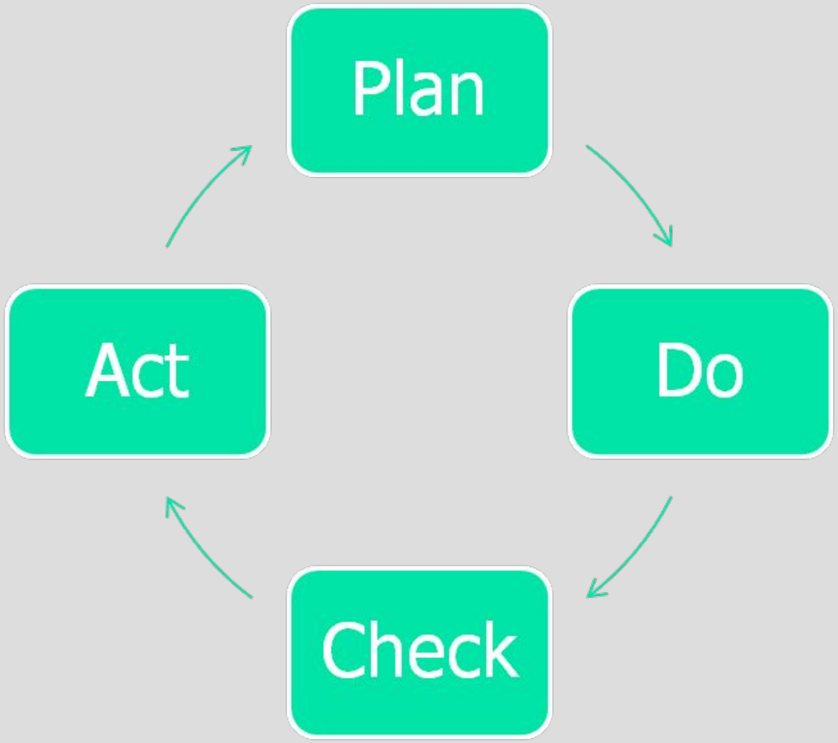
Project Targets



Characteristics of Project

- One time activity with quantifiable end results
- Existance of life cycle in project duration
- Set of Constraints
- Interdependance
- Uniqueness

Approach to Constant Improvement



Three Phases of PM

- Design it
- Do it
- Develop it

Project as a Conversion Process

Constraints
Financial, Legal, Ethical, Environmental and so on

Input

- Energy
- Resource

Project

- Mechanism
  - People
  - Know-how
  - Capital
  - Technology

Output

- Satisfaction

What is Project Management?

Project + Management

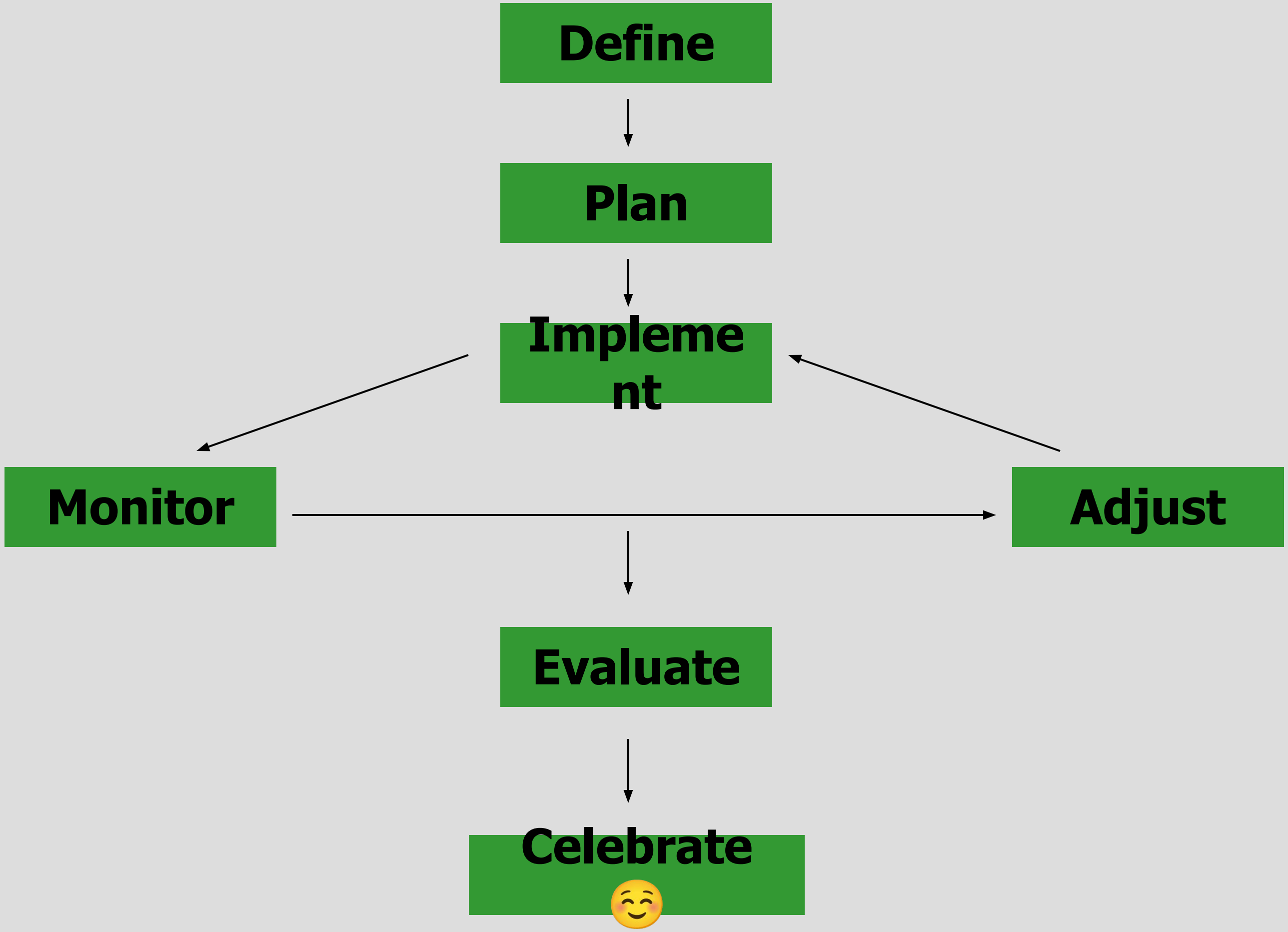
Specific Goal
Specific Resources
Specific Time Frame

Planning, Organizing
Staffing, Leading,
Controlling towards
goal attainment

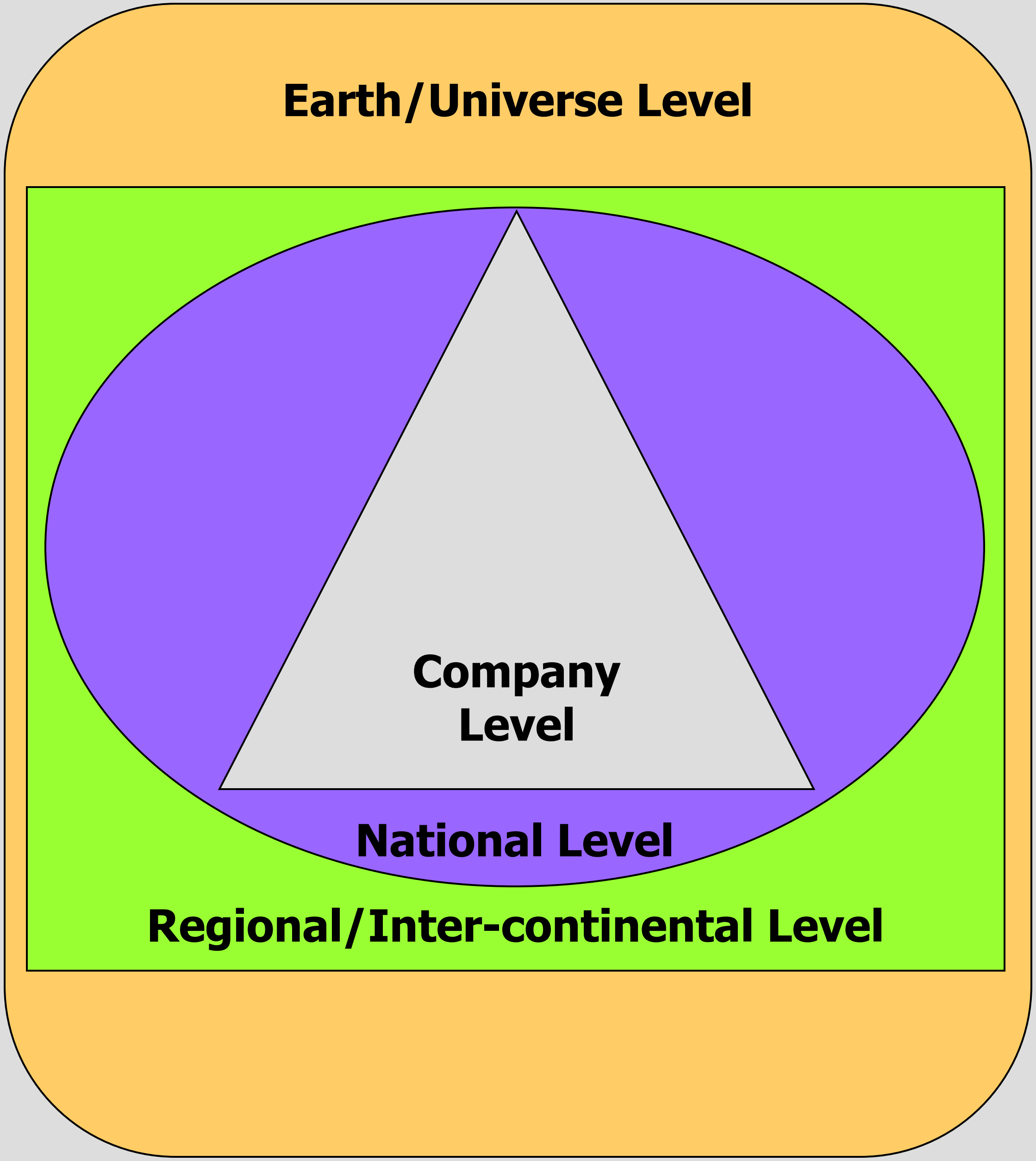
Development of Project Management

- Expansion of knowledge and more segregation in academic discipline
- Complex demand for goods and services
- Globalization
  - Cultural
  - Environmental

Stages of Project Management



Project Management Coverage

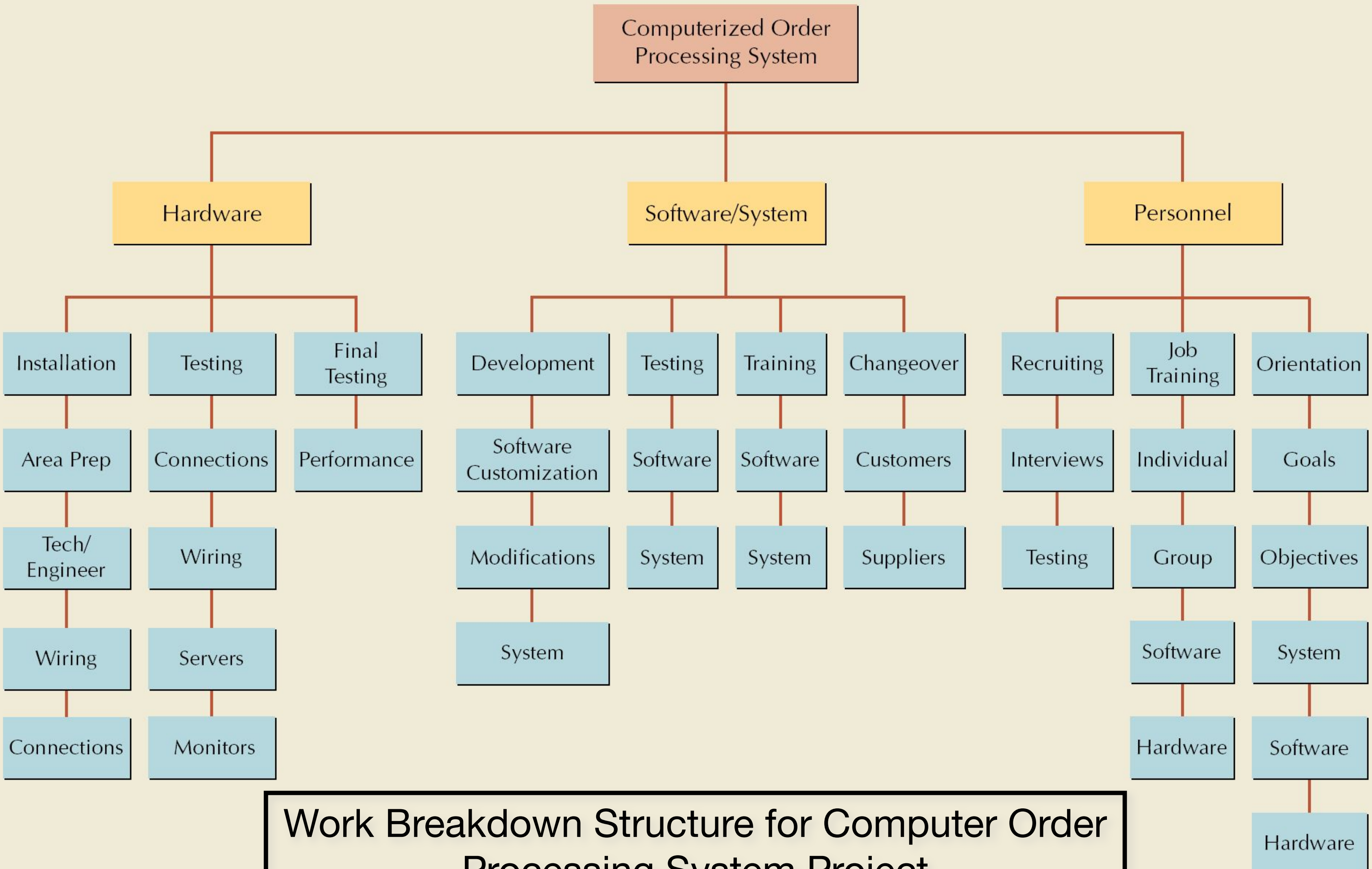


Project Management is the tool to achieve Goals at any level-

By initiating task, action and activity

Work breakdown structure

- A method of breaking down a project into individual elements (components, subcomponents, activities and tasks) in a hierarchical structure which can be scheduled
- It defines tasks that can be completed independently of other tasks, facilitating resource allocation, assignment of responsibilities and measurement and control of the project
- It is foundation of project planning
- It is developed before identification of dependencies and estimation of activity durations
- It can be used to identify the tasks in the CPM and PERT



Work Breakdown Structure for Computer Order Processing System Project

Project Scheduling and Control Techniques

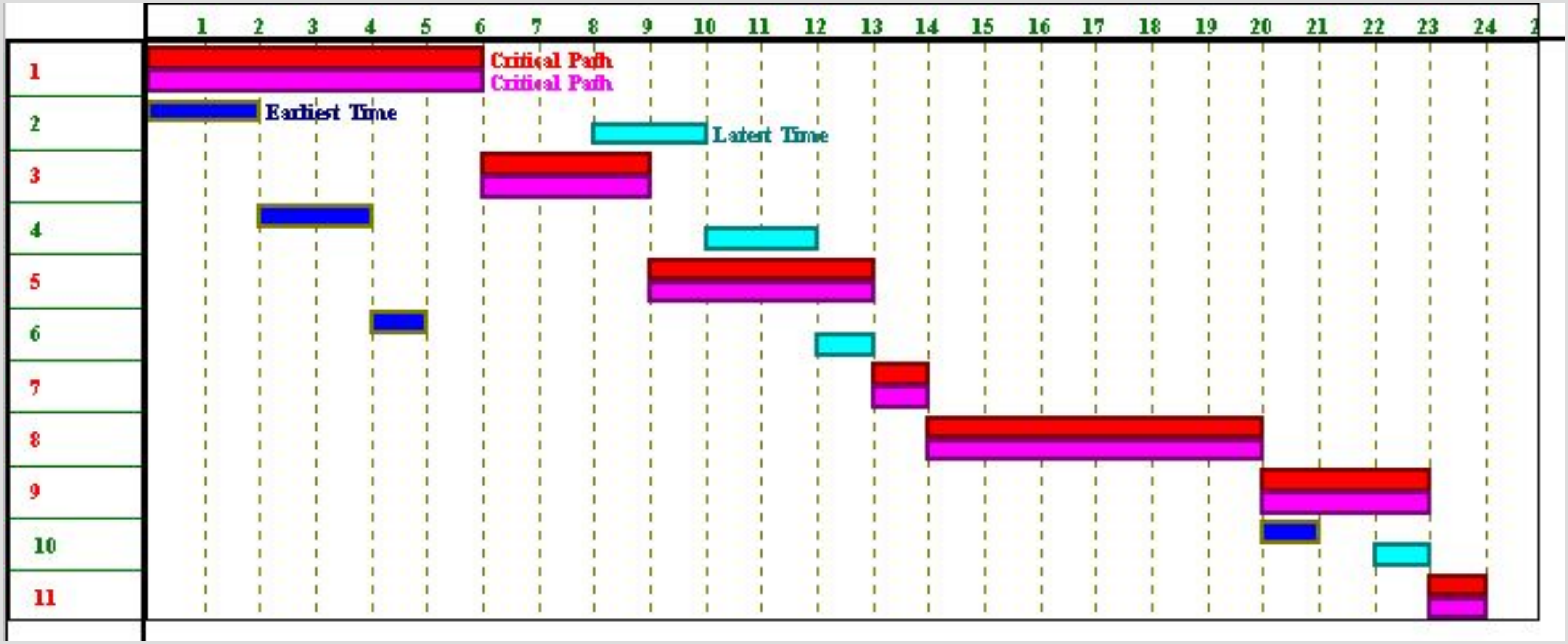
Gantt Chart

Critical Path Method (CPM)

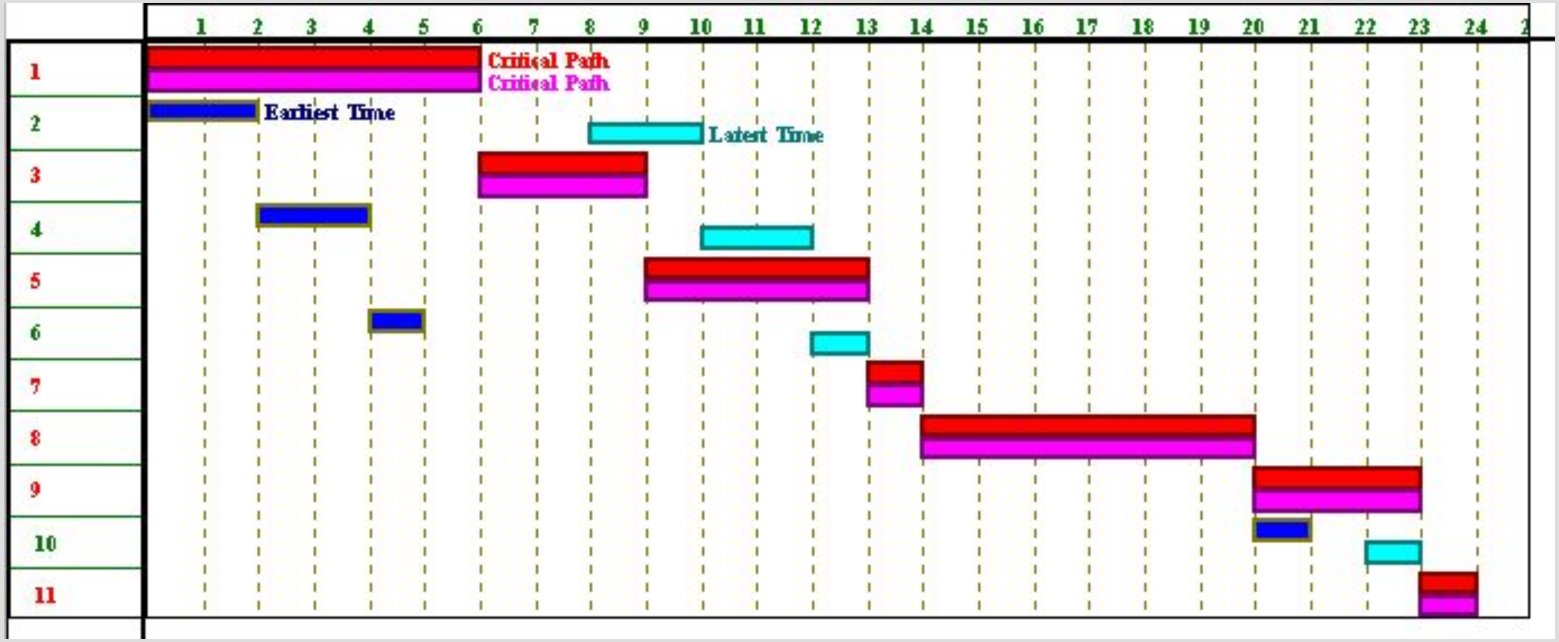
Program Evaluation and Review Technique (PERT)

Gantt Chart

- Graph or bar chart with a bar for each project activity that shows passage of time
- Provides visual display of project schedule



Gantt chart



Advantages

- Gantt charts are quite commonly used. They provide an easy graphical representation of when activities (might) take place.

Limitations

- Do not clearly indicate details regarding the progress of activities
- Do not give a clear indication of interrelation ship between the separate activities

CPM - Critical Path Method

- Definition: In **CPM** activities are shown as a network of precedence relationships using activity-on-node network construction
  - Single estimate of activity time
  - Deterministic activity times

USED IN : Production management - for the jobs of repetitive in nature where the activity time estimates can be predicted with considerable certainty due to the existence of past experience.

PERT -
Project Evaluation & Review Techniques

- Definition: In **PERT** activities are shown as a network of precedence relationships using activity-on-arrow network construction
  – Multiple time estimates
  – Probabilistic activity times

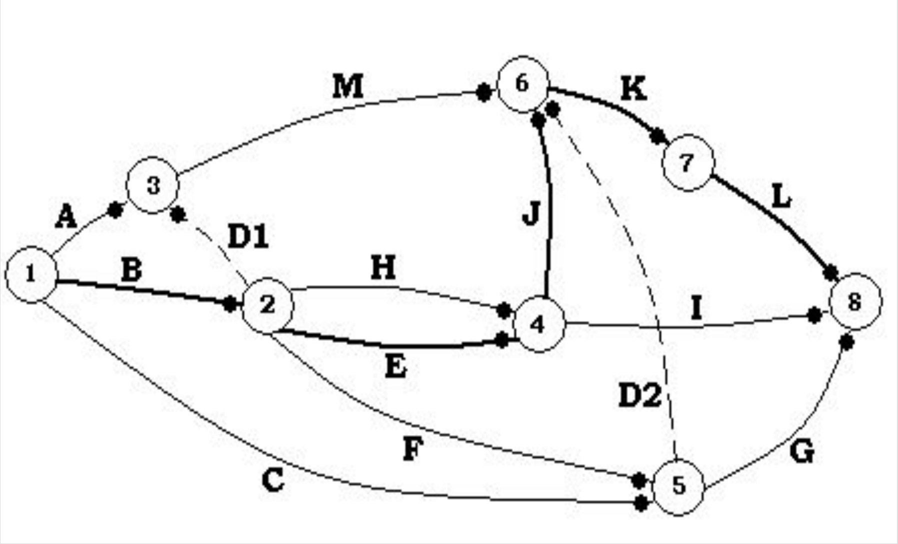
USED IN : Project management - for non-repetitive jobs (research and development work), where the time and cost estimates tend to be quite uncertain. This technique uses probabilistic time estimates.

CPM/PERT

- Deficiencies (in Gantt Chart) can be eliminated to a large extent by showing the interdependence of various activities by means of connecting arrows called network technique.

- Overtime CPM and PERT became one technique

- ADVANTAGES:
  - Precedence relationships
  - large projects
  - more efficient



Project Network

- Network analysis is the general name given to certain specific techniques which can be used for the planning, management and control of projects
- Use of nodes and arrows

  Arrows An arrow leads from tail to head directionally

  – Indicate ACTIVITY, a time consuming effort that is required to perform a part of the work.

  Nodes A node is represented by a circle

  - Indicate EVENT, a point in time where one or more activities start and/or finish.
- Activity
  – A task or a certain amount of work required in the project
  – Requires time to complete
  – Represented by an arrow
- Dummy Activity
  – Indicates only precedence relationships
  – Does not require any time of effort

Project Network

- Event
  - Signals the beginning or ending of an activity
  - Designates a point in time
  - Represented by a circle (node)
- Network
  - Shows the sequential relationships among activities using nodes and arrows

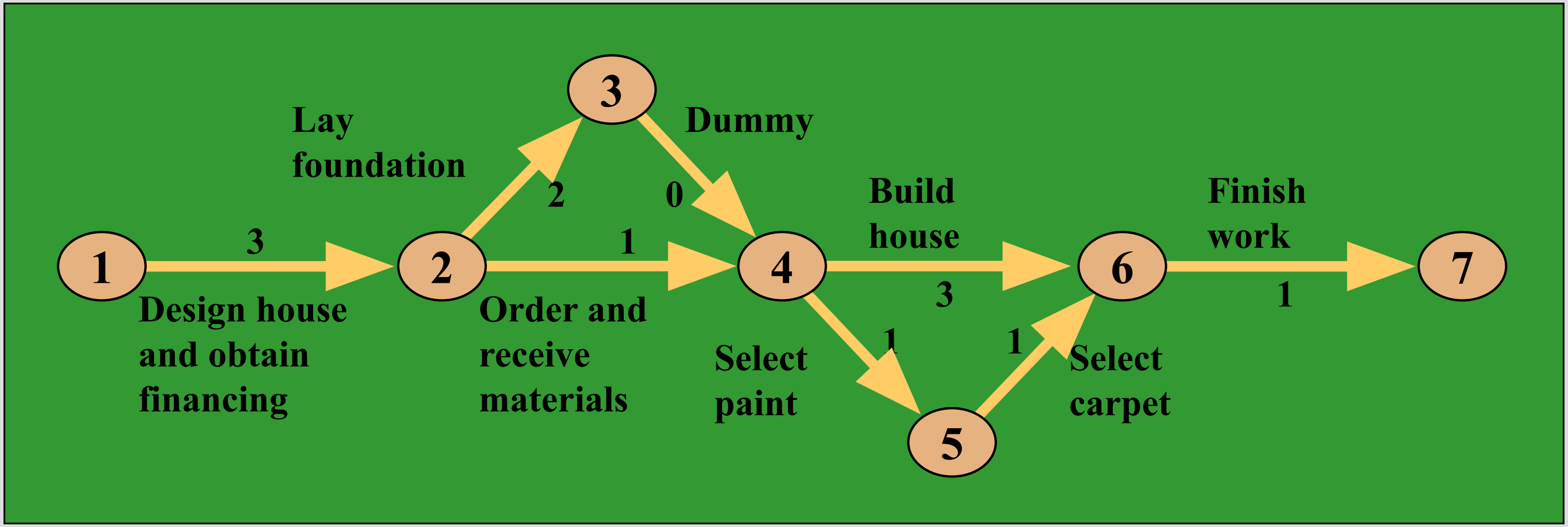
◆Activity-on-node (AON)

nodes represent activities, and arrows show precedence relationships

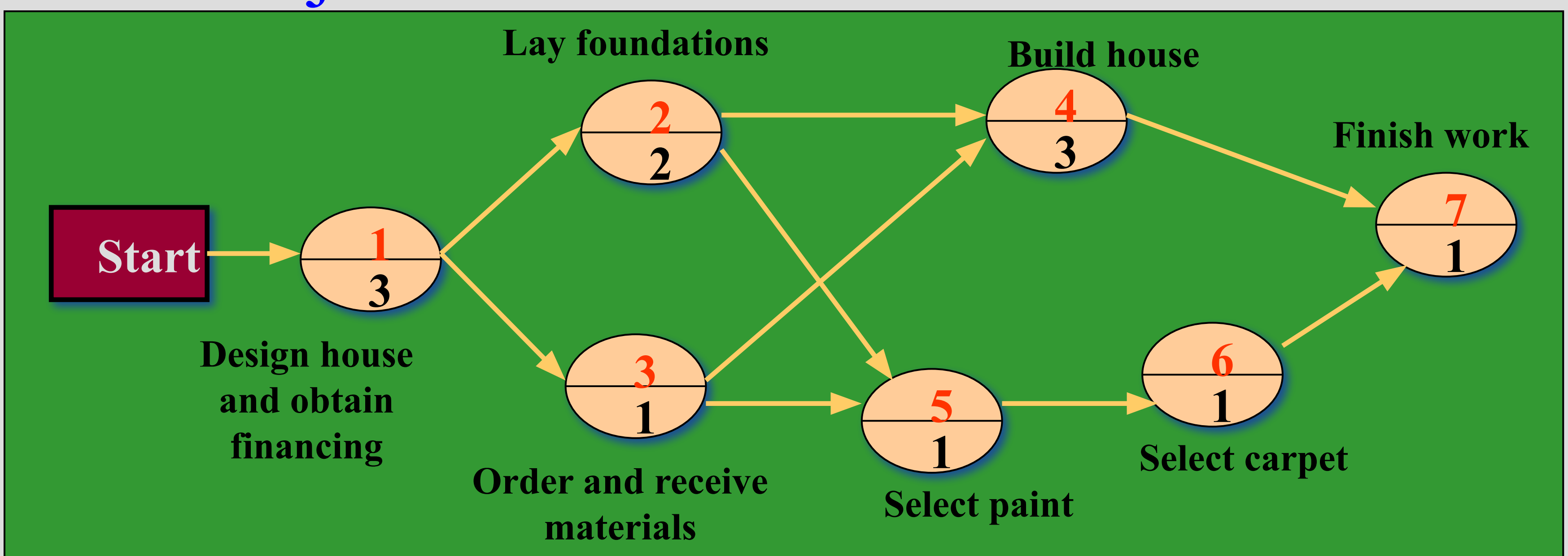
◆Activity-on-arrow (AOA)

arrows represent activities and nodes are events for points in time

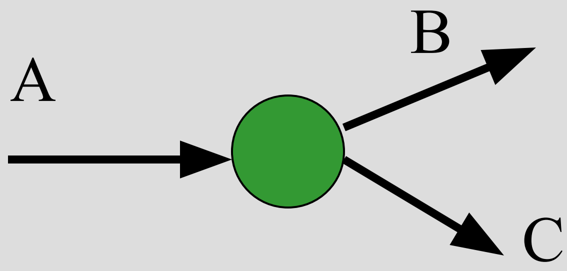
AOA Project Network for House



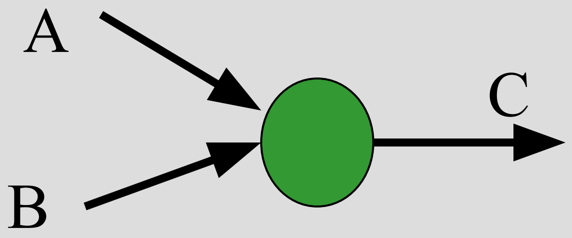
AON Project Network for House



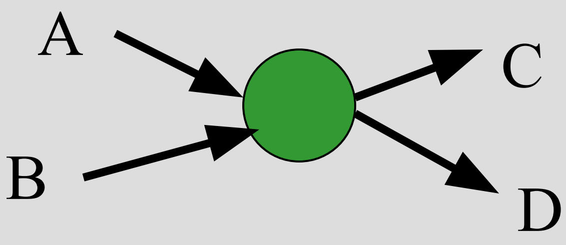
Situations in network diagram



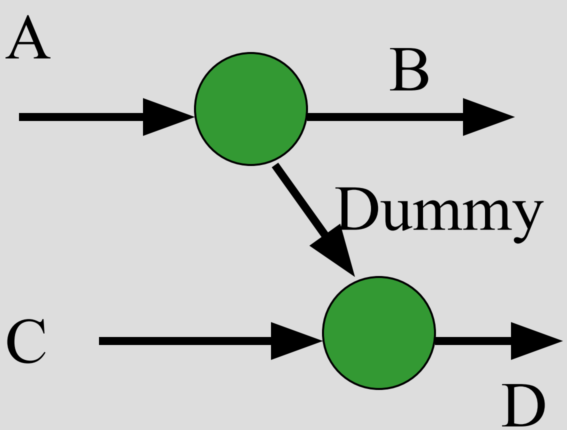
A must finish before either B or C can start



both A and B must finish before C can start



both A and C must finish before either of B or D can start



A must finish before B can start

both A and C must finish before D can start

CPM calculation

- <u>Path</u>
  – A connected sequence of activities leading from the starting event to the ending event
- <u>Critical Path</u>
  – The longest path (time); determines the project duration
- <u>Critical Activities</u>
  – All of the activities that make up the critical path

Forward Pass

- Earliest Start Time (ES)
  - earliest time an activity can start
  - ES = maximum EF of immediate predecessors
- Earliest finish time (EF)
  - earliest time an activity can finish
  - earliest start time plus activity time

$$EF = ES + t$$

Backward Pass

- Latest Start Time (LS)

  Latest time an activity can start without delaying critical path time

  $$LS = LF - t$$

- Latest finish time (LF)

  latest time an activity can be completed without delaying critical path time

  LS = minimum LS of immediate predecessors

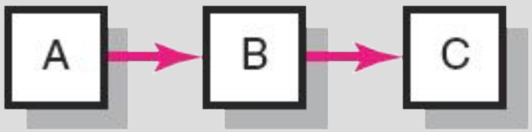
CPM analysis

- Draw the CPM network
- Analyze the paths through the network
- Determine the float for each activity
  - Compute the activity's float

    $$\text{float} = LS - ES = LF - EF$$

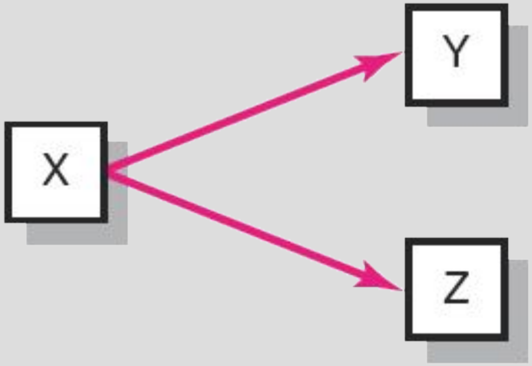
  - Float is the maximum amount of time that this activity can be delay in its completion before it becomes a critical activity, i.e., delays completion of the project
- Find the critical path is that the sequence of activities and events where there is no "slack" i.e.. Zero slack
  - Longest path through a network
- Find the project duration is minimum project completion time

Activity-on-Node Fundamentals



A is preceded by nothing
B is preceded by A
C is preceded by B

(A)

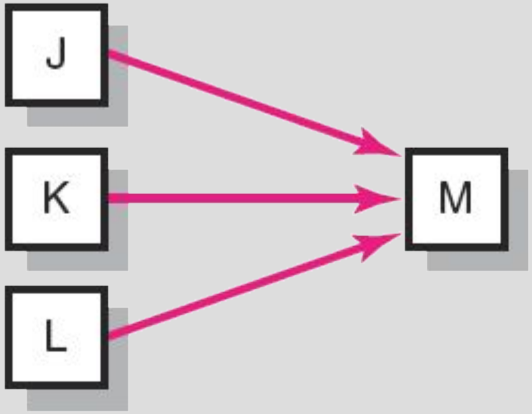


Y and Z are preceded by X

Y and Z can begin at the same time, if you wish

(B)

Activity-on-Node Fundamentals (cont'd)

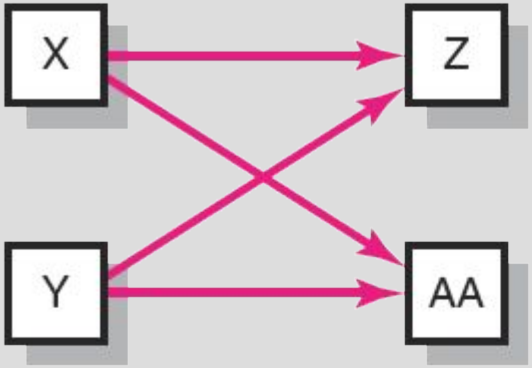


J, K, & L can all begin at the same time, if you wish (they need not occur simultaneously)

but

All (J, K, L) must be completed before M can begin

(C)



Z is preceded by X and Y

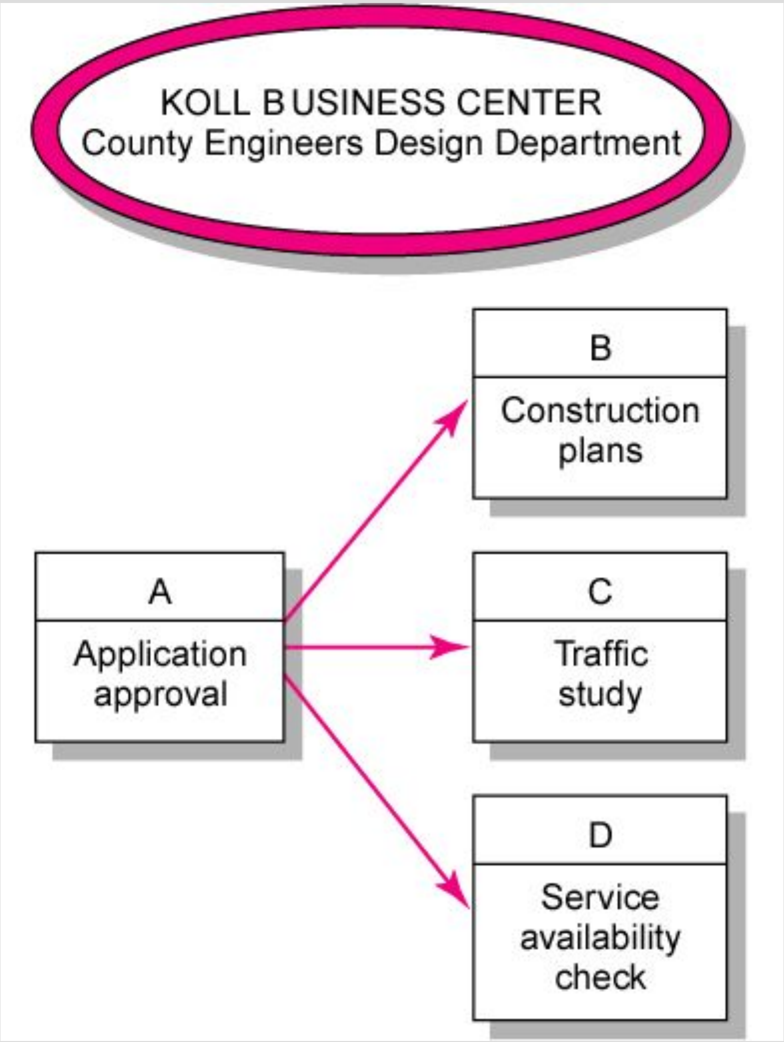
AA is preceded by X and Y

(D)

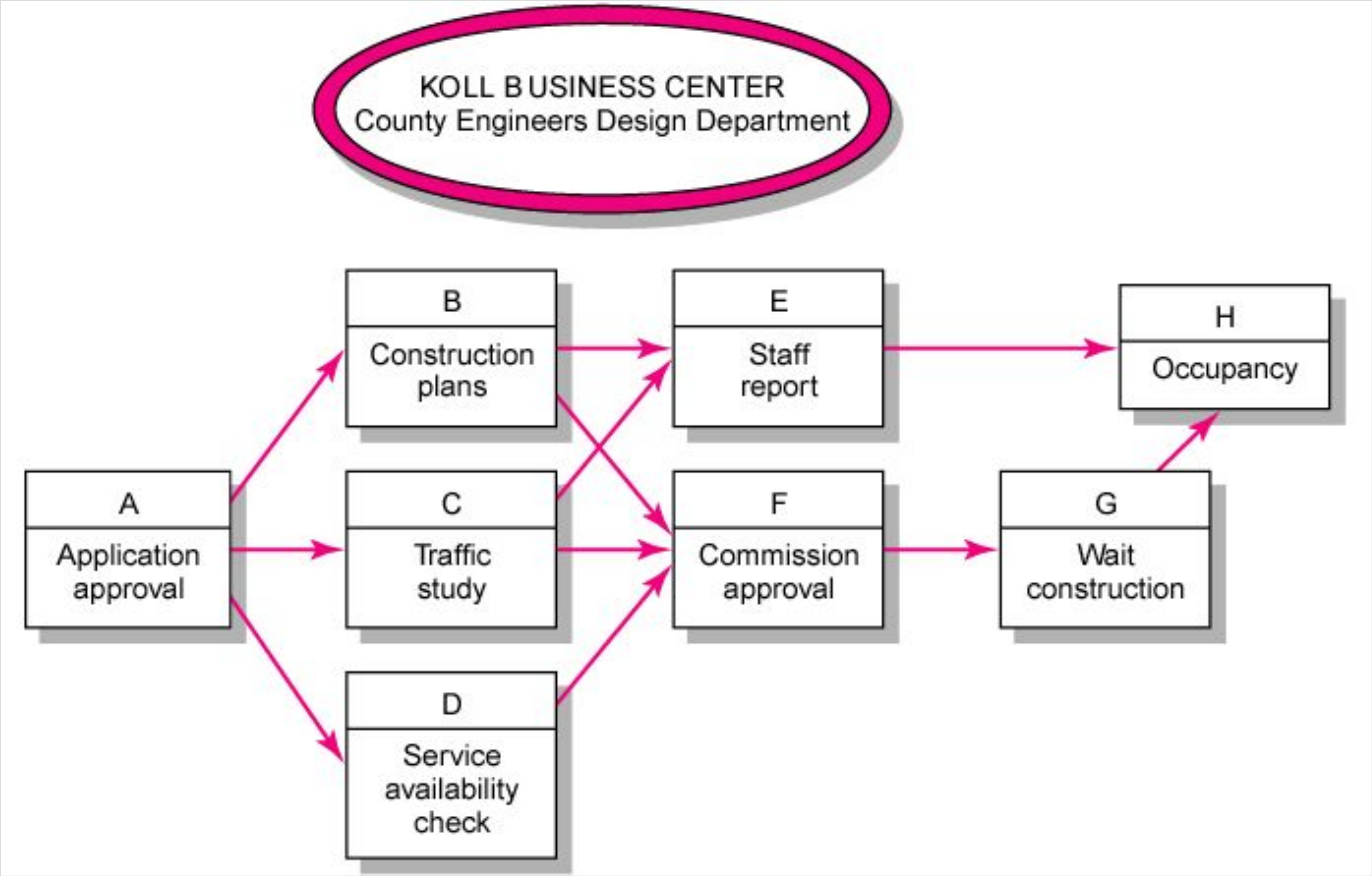
Network Information

KOLL BUSINESS CENTER County Engineers Design Department		
Activity	**Description**	**Preceding Activity**
A	Application approval	None
B	Construction plans	A
C	Traffic study	A
D	Service availability check	A
E	Staff report	B, C
F	Commission approval	B, C, D
G	Wait for construction	F
H	Occupancy	E, G

Koll Business Center—Partial Network



Koll Business Center—Complete Network



Network Computation Process

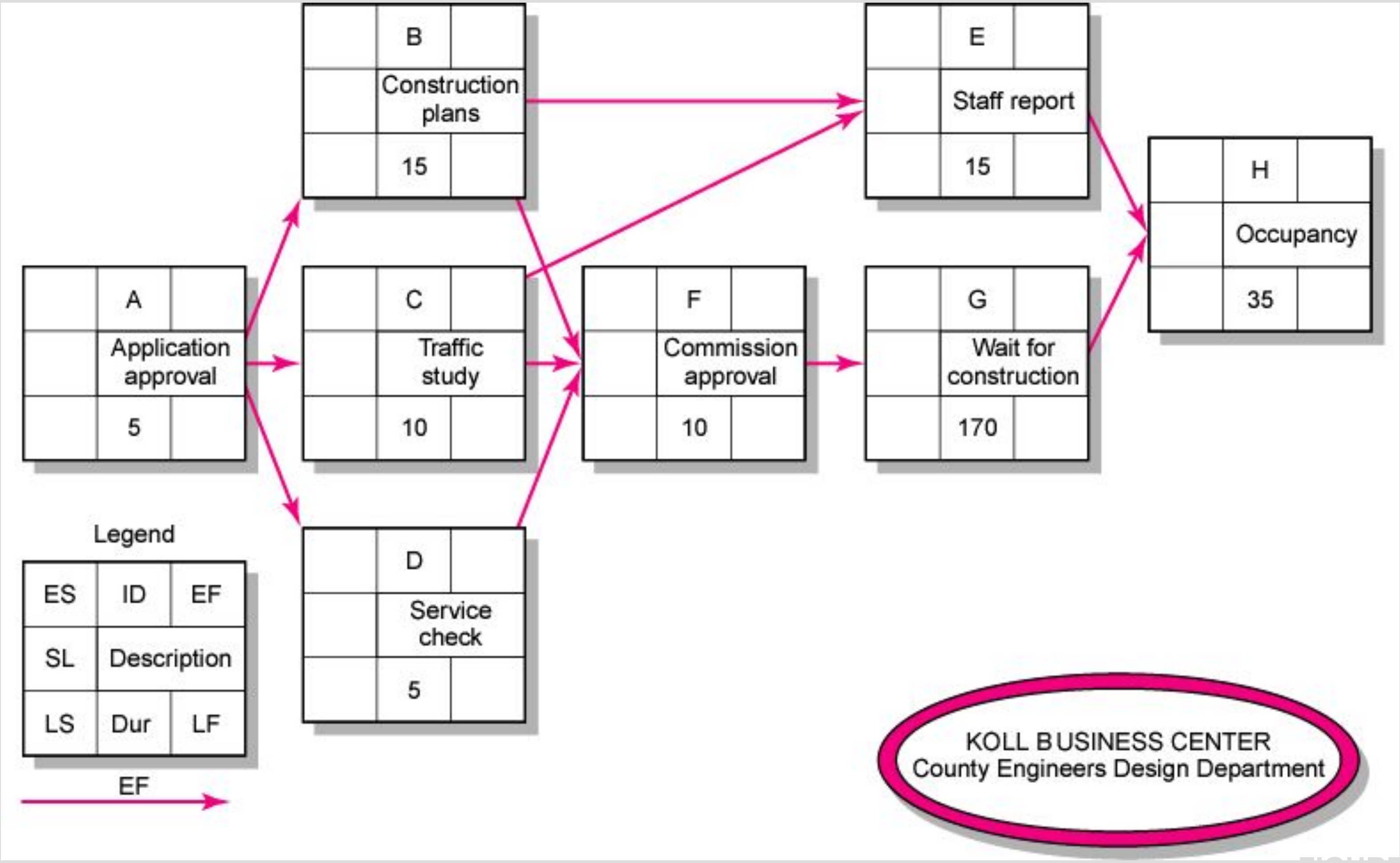
- Forward Pass—Earliest Times
  - How soon can the activity start? (early start—ES)
  - How soon can the activity finish? (early finish—EF)
  - How soon can the project finish? (expected time—ET)
- Backward Pass—Latest Times
  - How late can the activity start? (late start—LS)
  - How late can the activity finish? (late finish—LF)
  - Which activities represent the critical path?
  - How long can it be delayed? (slack or float—SL)

Network Information

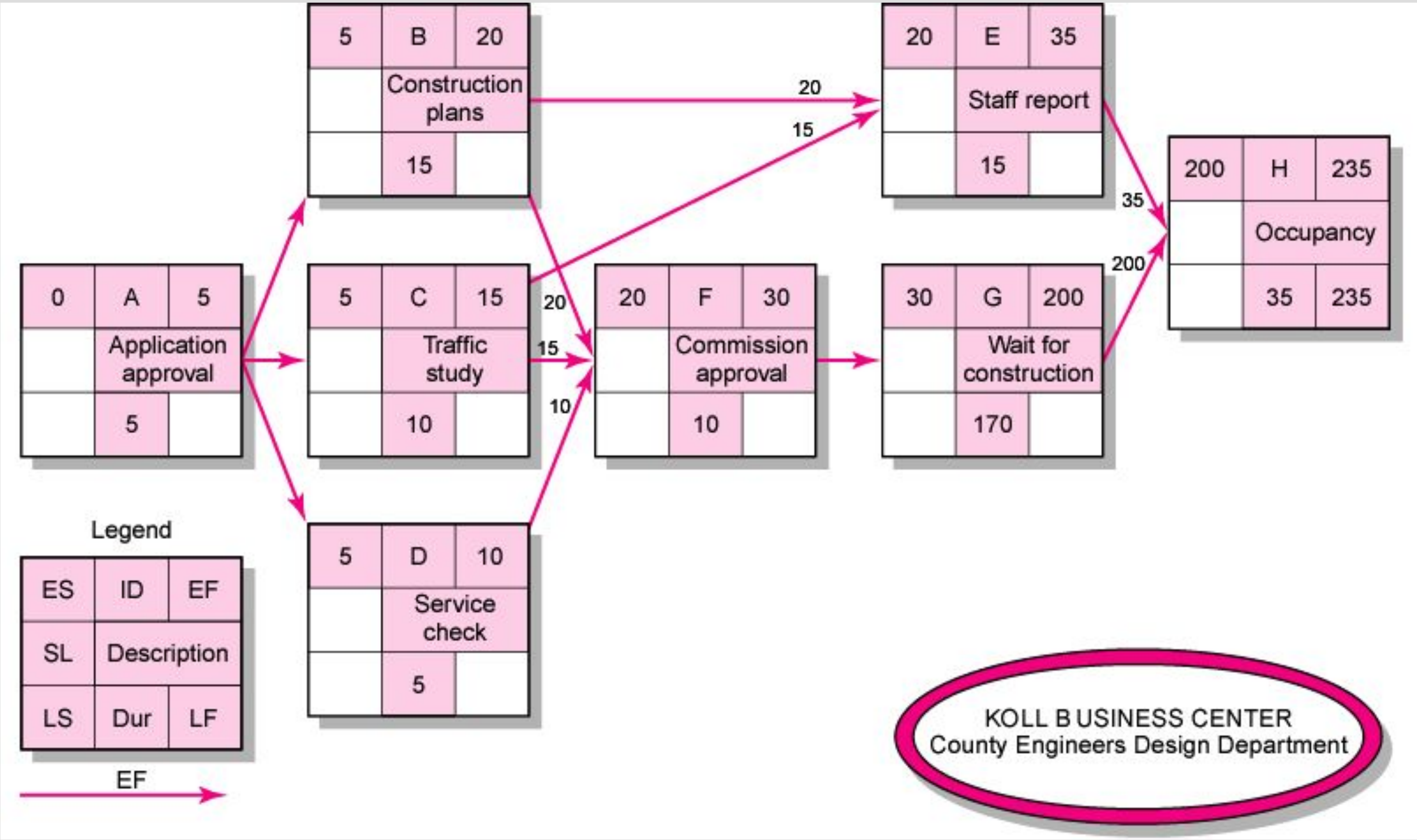
KOLL BUSINESS CENTER
County Engineers Design Department

Activity	Description	Preceding Activity	Activity Time
A	Application approval	None	5
B	Construction plans	A	15
C	Traffic study	A	10
D	Service availability check	A	5
E	Staff report	B, C	15
F	Commission approval	B, C, D	10
G	Wait for construction	F	170
H	Occupancy	E, G	35

Activity-on-Node Network



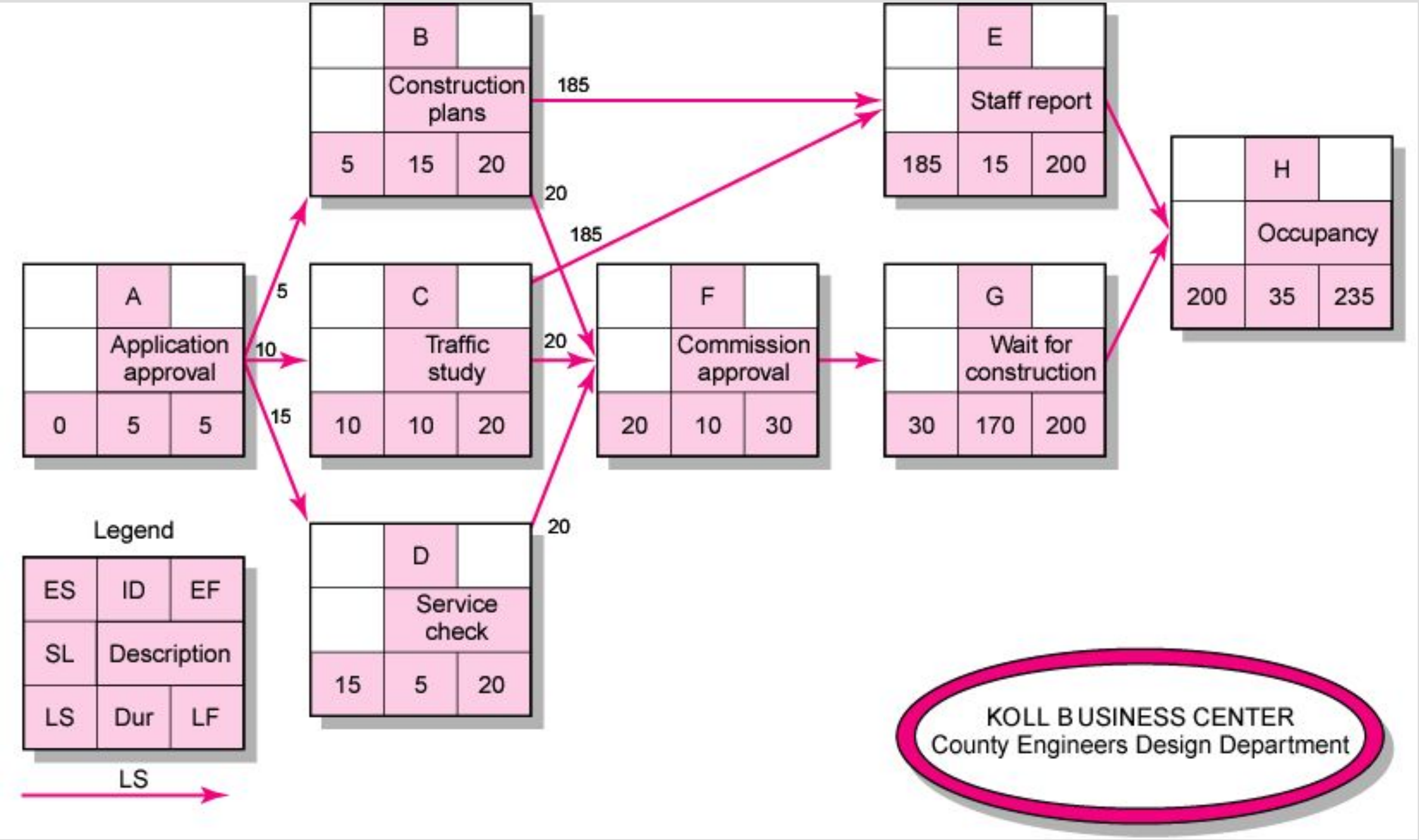
Activity-on-Node Network Forward Pass



Forward Pass Computation

- Add activity times along each path in the network (ES + Duration = EF).
- Carry the early finish (EF) to the next activity where it becomes its early start (ES) ***unless…***
- The next succeeding activity is a merge activity, in which case the largest EF of all preceding activities is selected.

Activity-on-Node Network Backward Pass



Backward Pass Computation

- Subtract activity times along each path in the network (LF - Duration = LS).
- Carry the late start (LS) to the next activity where it becomes its late finish (LF) ***unless...***
- The next succeeding activity is a burst activity, in which case the smallest LF of all preceding activities is selected.

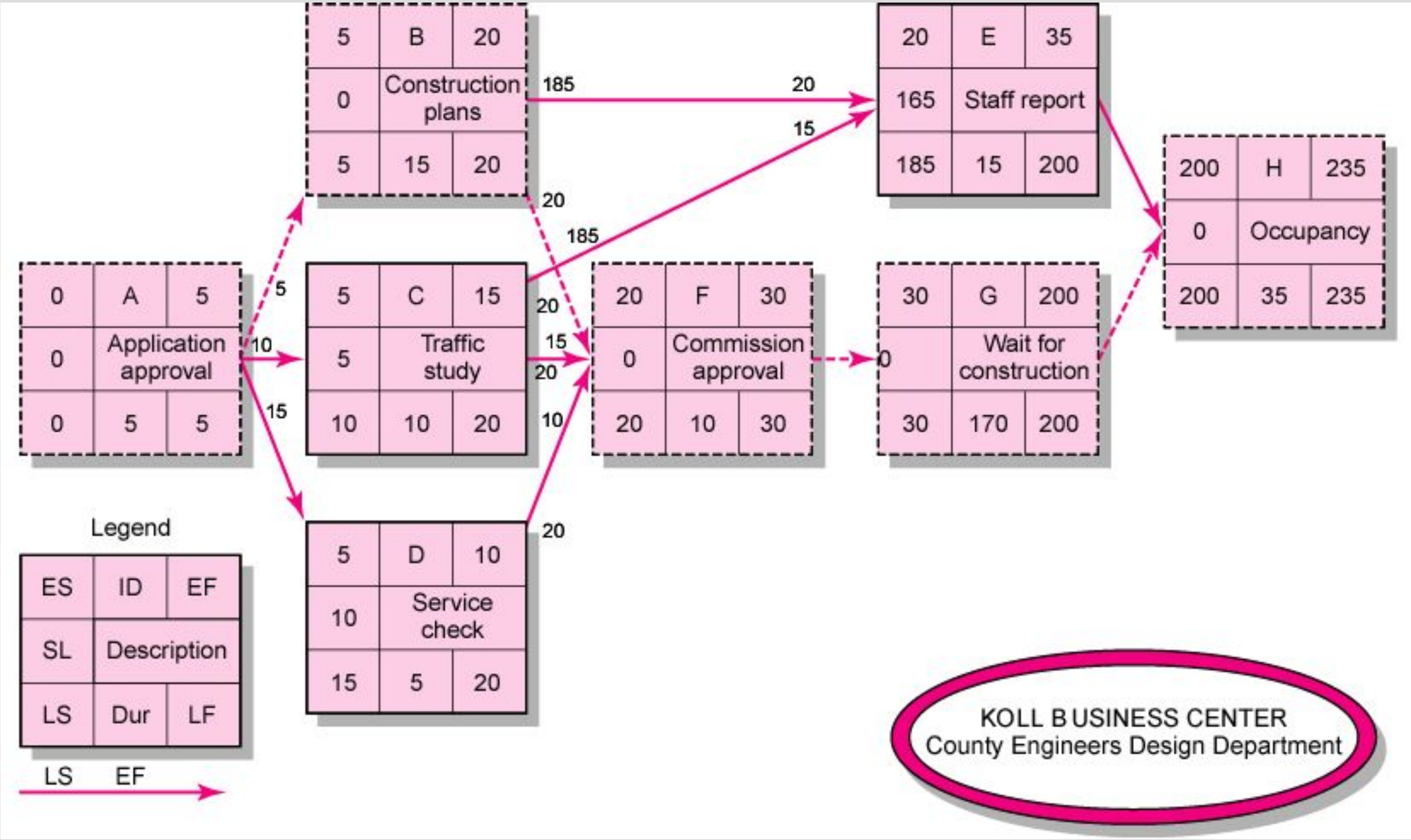
Determining Slack (or Float)

- Free Slack (or Float)
  - The amount of time an activity can be delayed without delaying connected successor activities
- Total Slack
  - The amount of time an activity can be delayed without delaying the entire project
- The critical path is the network path(s) that has (have) the least slack in common.

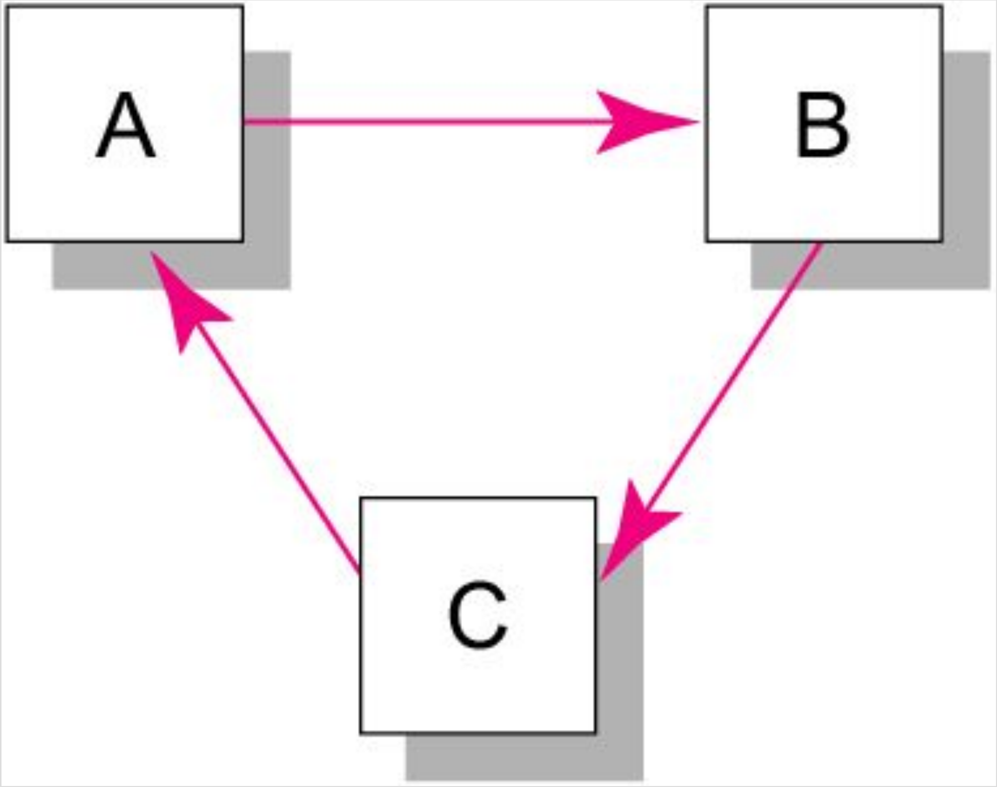
Sensitivity of a Network

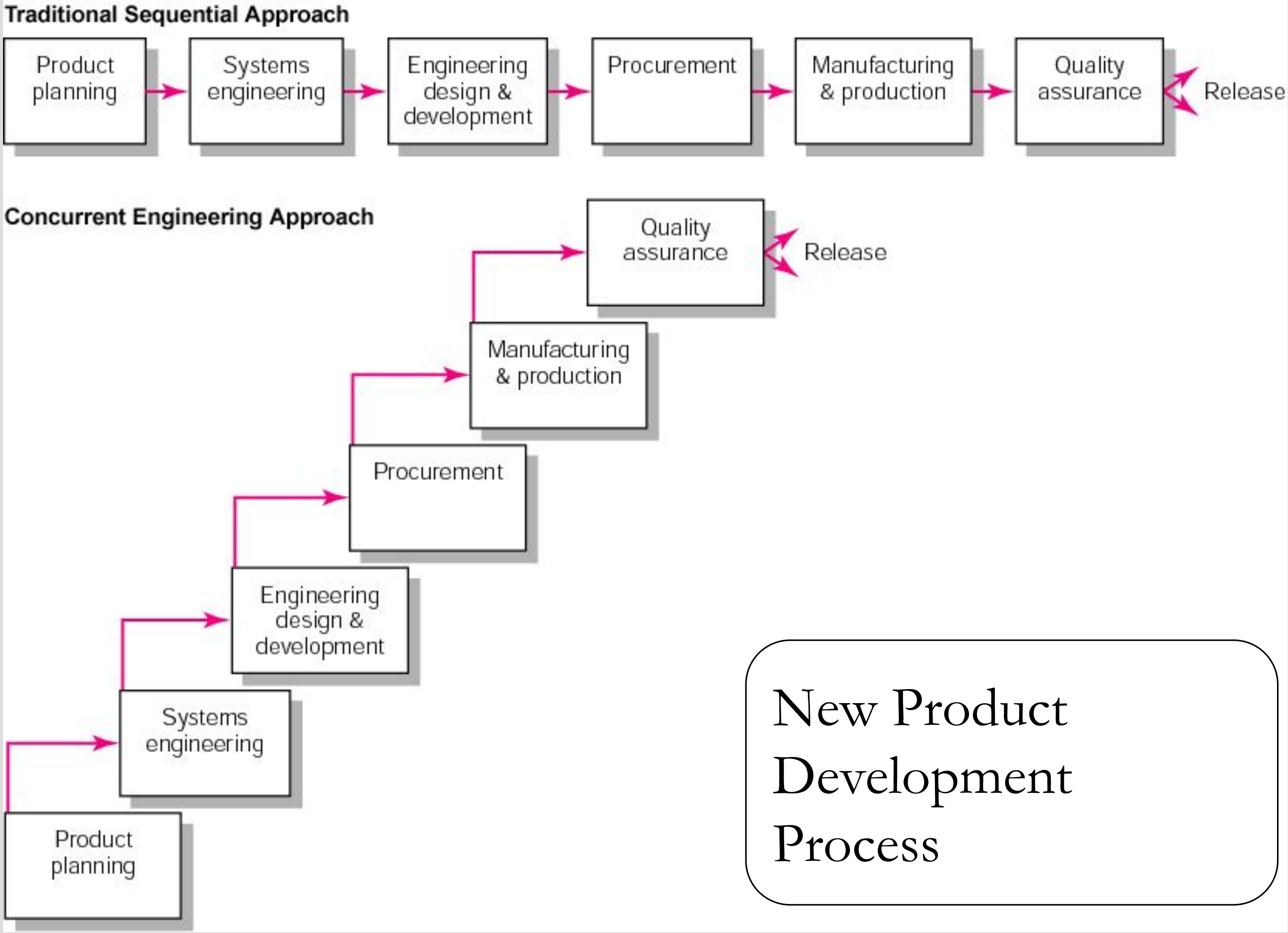
- The likelihood the original critical path(s) will change once the project is initiated.
  - **Function of:**
    - **The number of critical paths**
    - **The amount of slack across near critical activities**

Activity-on-Node Network with Slack



Illogical Loop





New Product Development Process

Key Terms

Activity

Activity-on-arrow (AOA)

Activity-on-node (AON)

Concurrent engineering

Critical path

Early and late times

Gantt chart

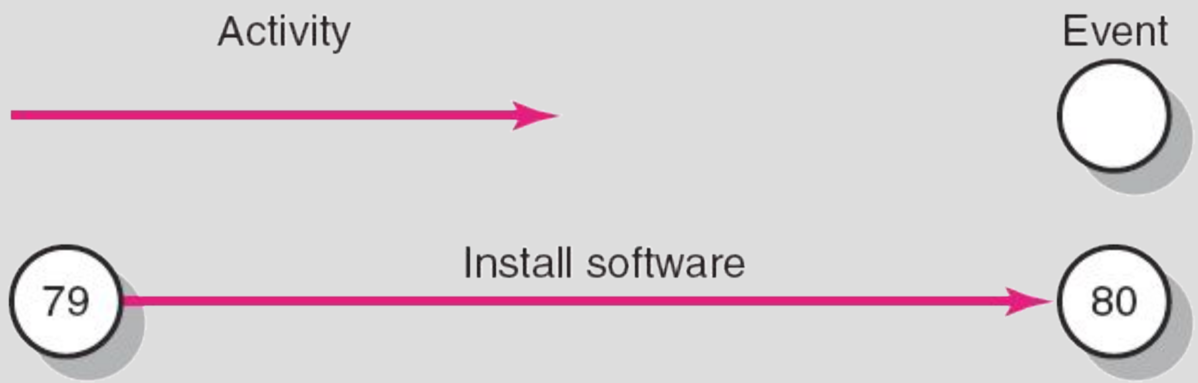
Merge activity

Network sensitivity

Parallel activity

Slack/float

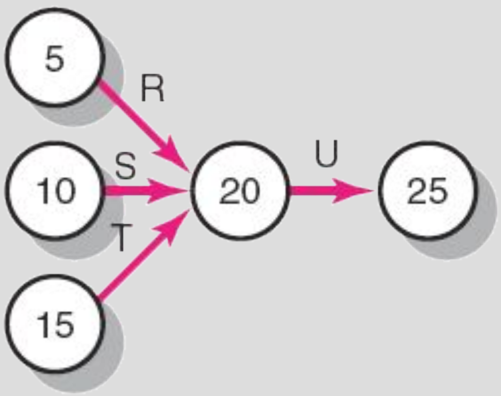
Activity-on-Arrow Network Building Blocks



Activity-on-Arrow Network Fundamentals

Y is preceded by X

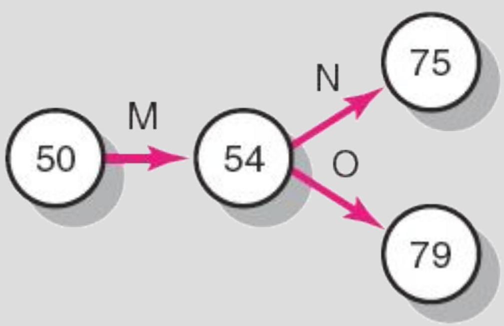
(A)



U is preceded by R, S, T

R, S, T can occur concurrently, if you wish

(B)

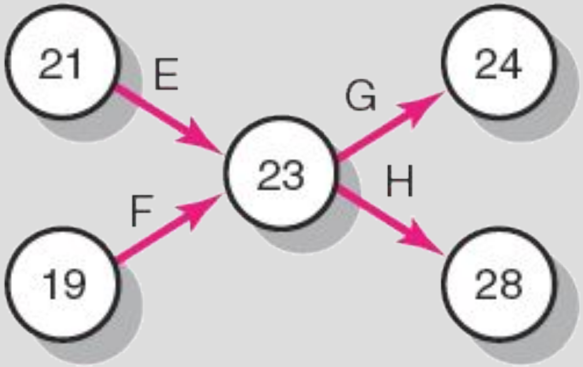


N and O are preceded by M

When M is complete, N and O can occur concurrently, if you wish

(C)

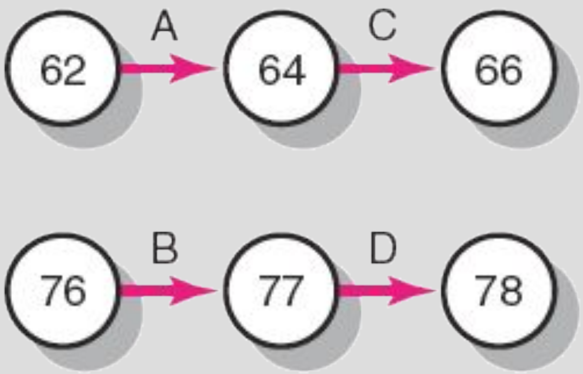
Activity-on-Arrow Network Fundamentals



E and F must precede G and H

E and F can occur together, if you wish
G and H can occur together, if you wish

(D)



A must precede C
B must precede D

Path A–C is independent of path B–D

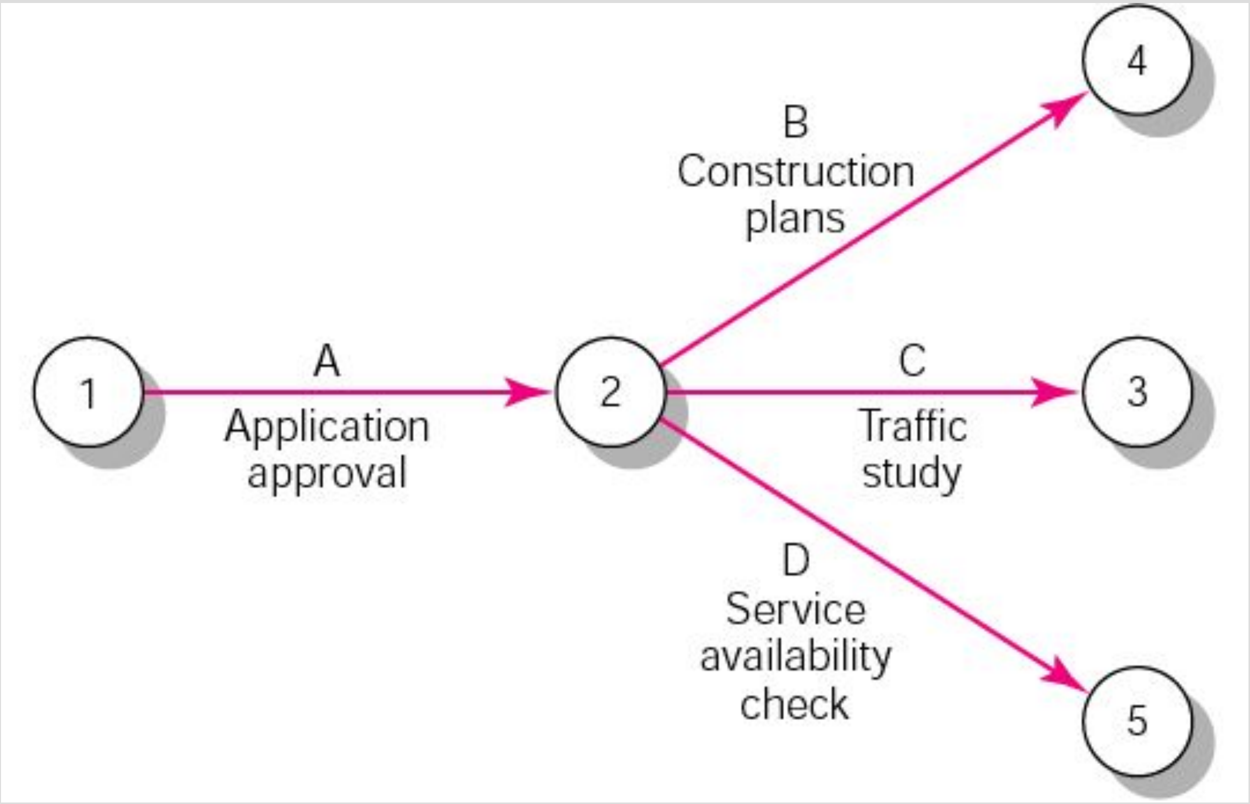
(E)

Koll Center Project: Network Information

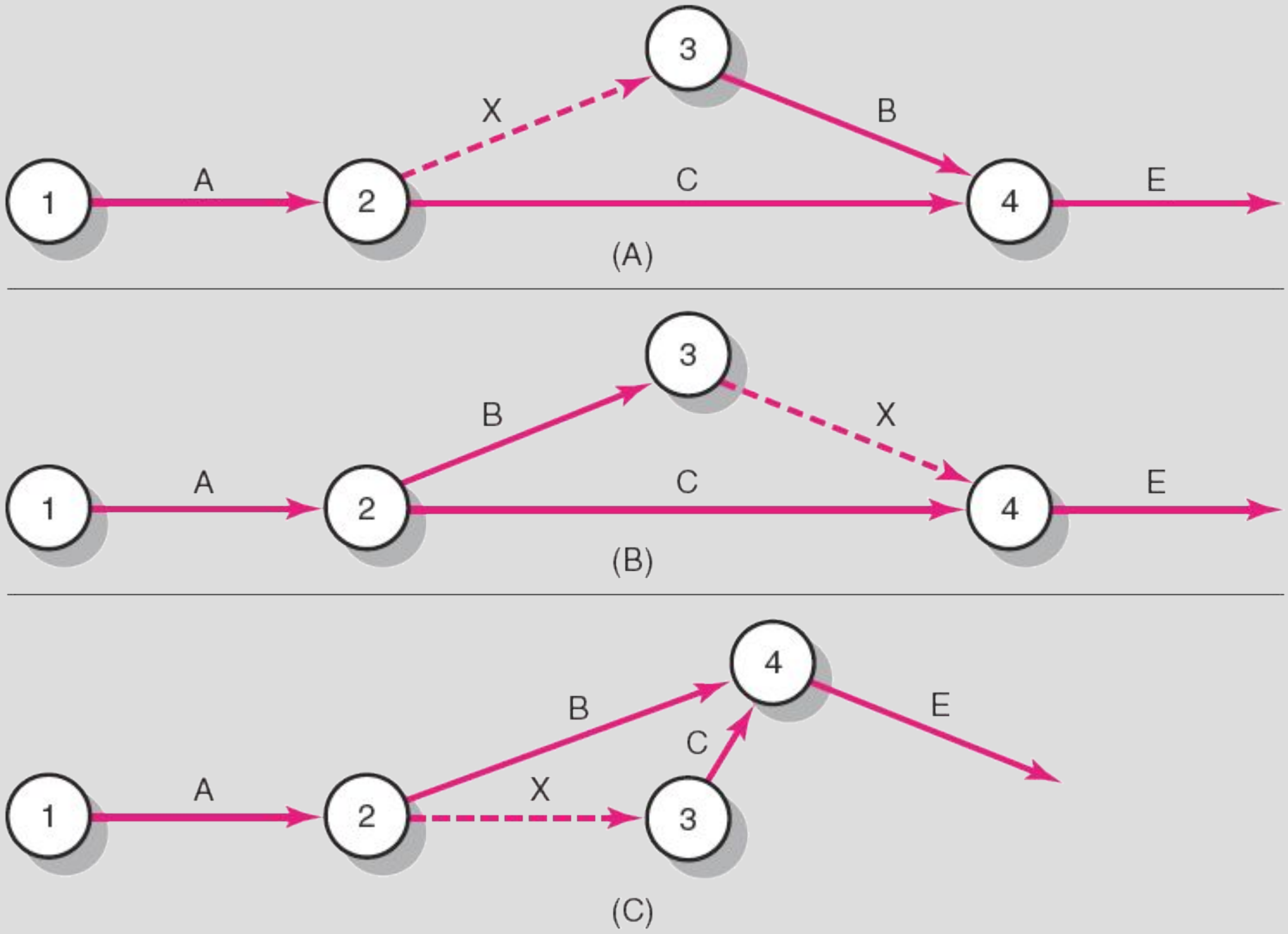
KOLL BUSINESS CENTER
County Engineers Design Department

Activity	Description	Preceding Activity	Activity Time
A	Application approval	None	5
B	Construction plans	A	15
C	Traffic study	A	10
D	Service availability check	A	5
E	Staff report	B, C	15
F	Commission approval	B, C, D	10
G	Wait for construction	F	170
H	Occupancy	E, G	35

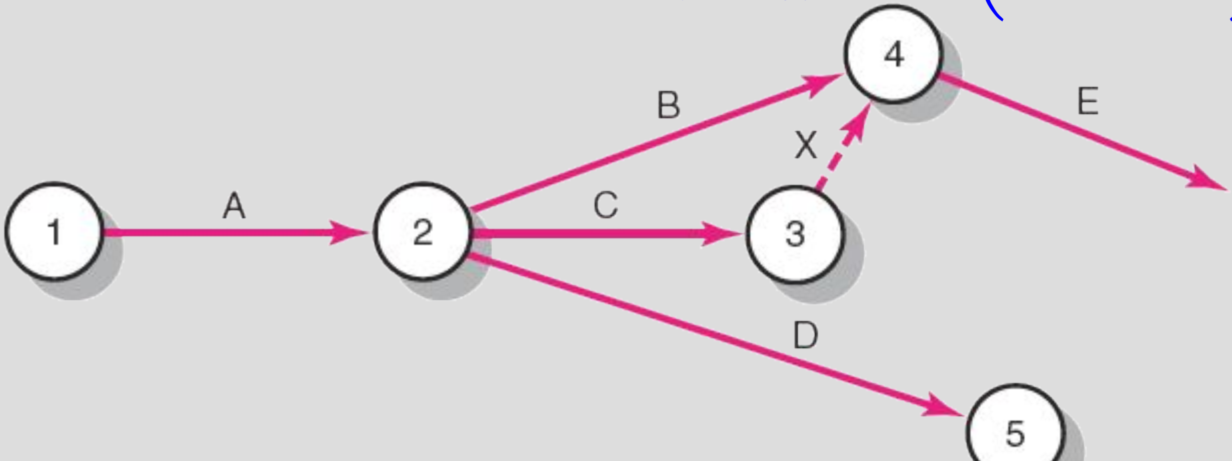
Partial Koll Business Center AOA Network



Partial AOA Koll Network



Partial AOA Koll Network (cont'd)



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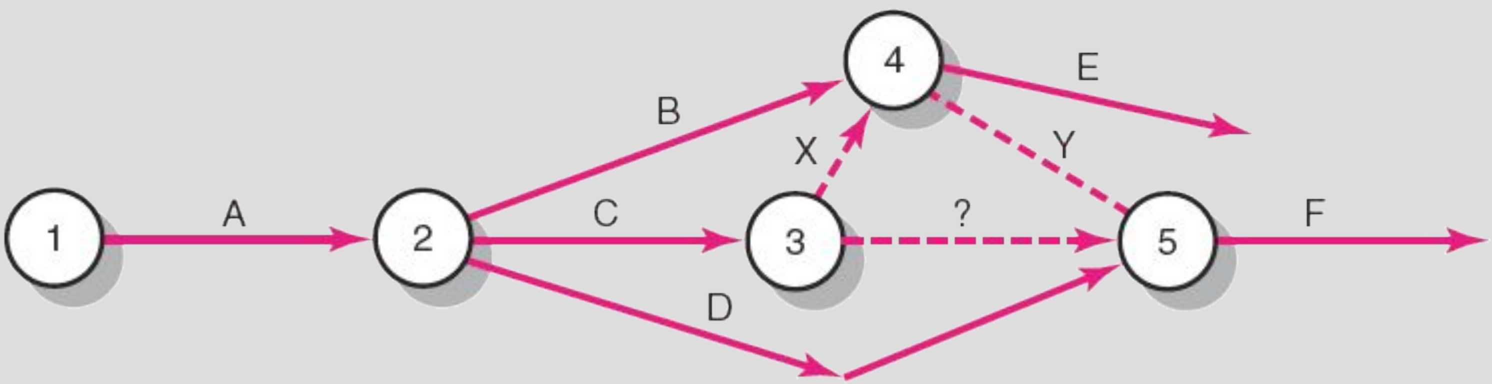
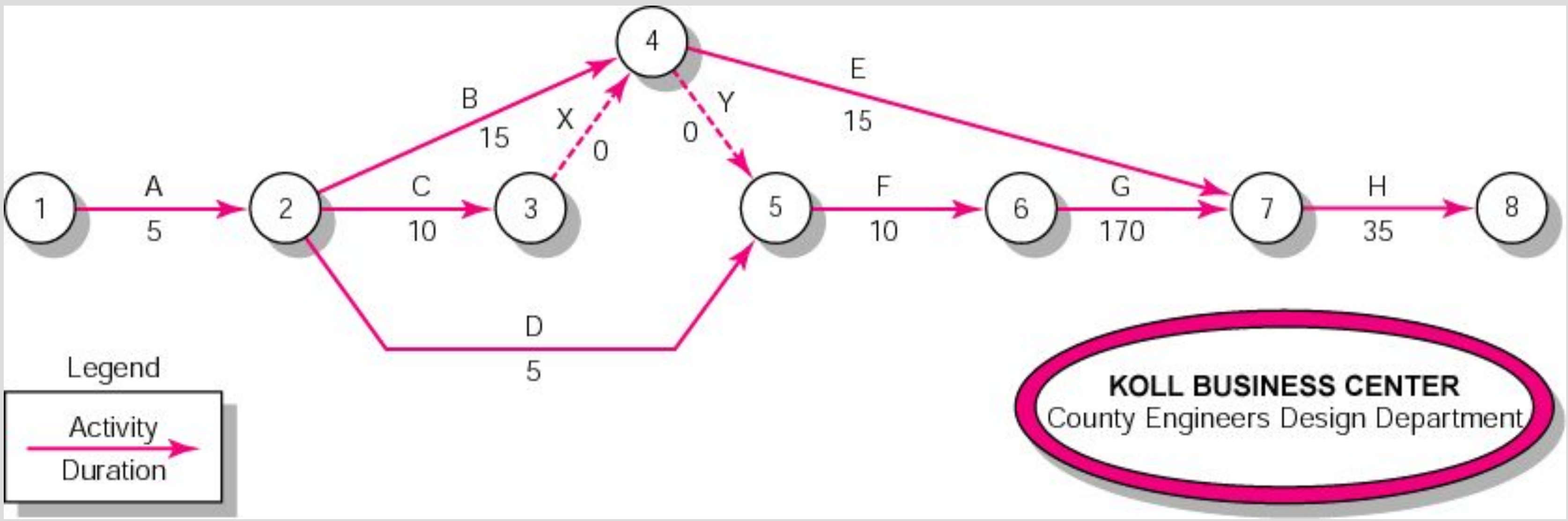
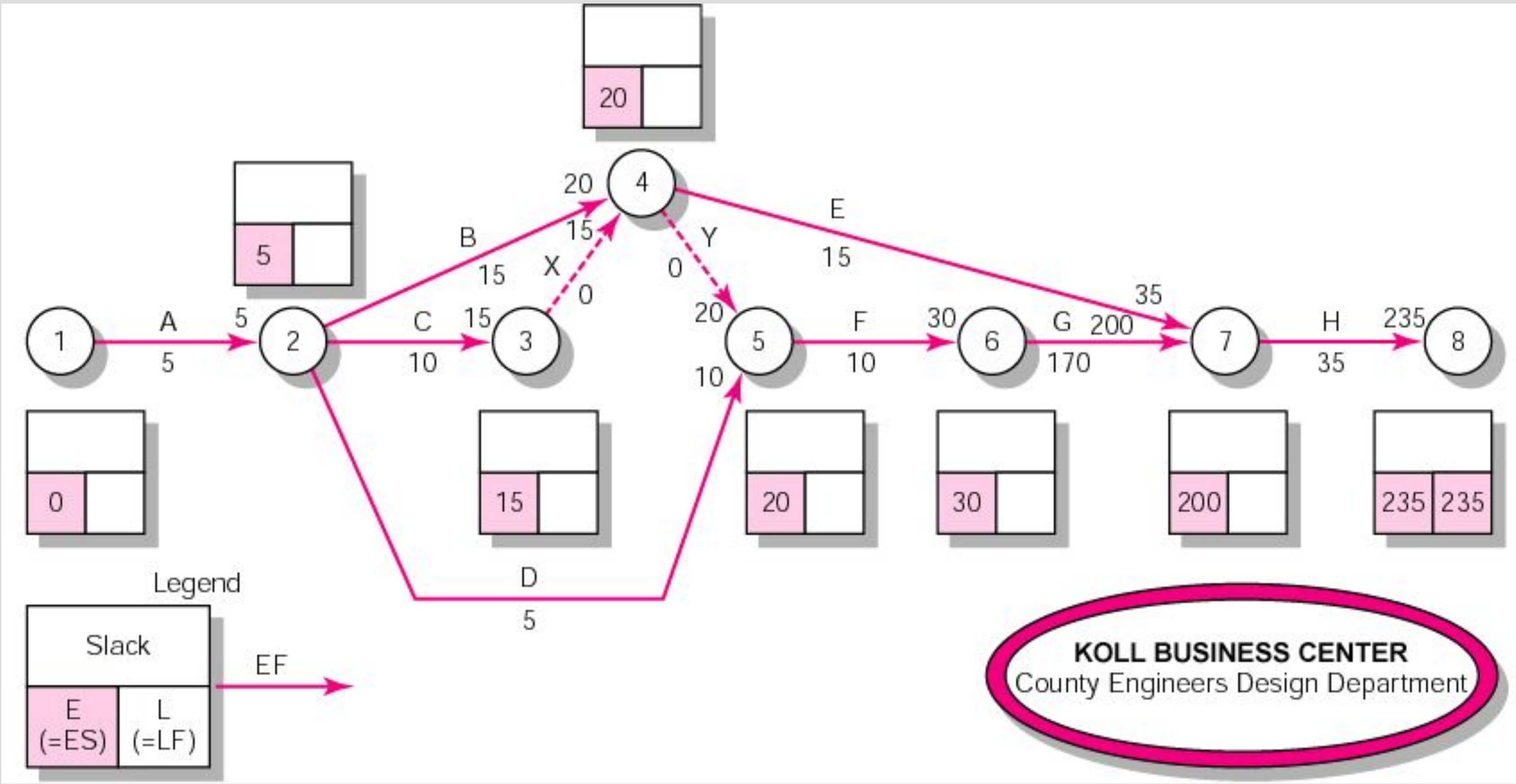


(E)

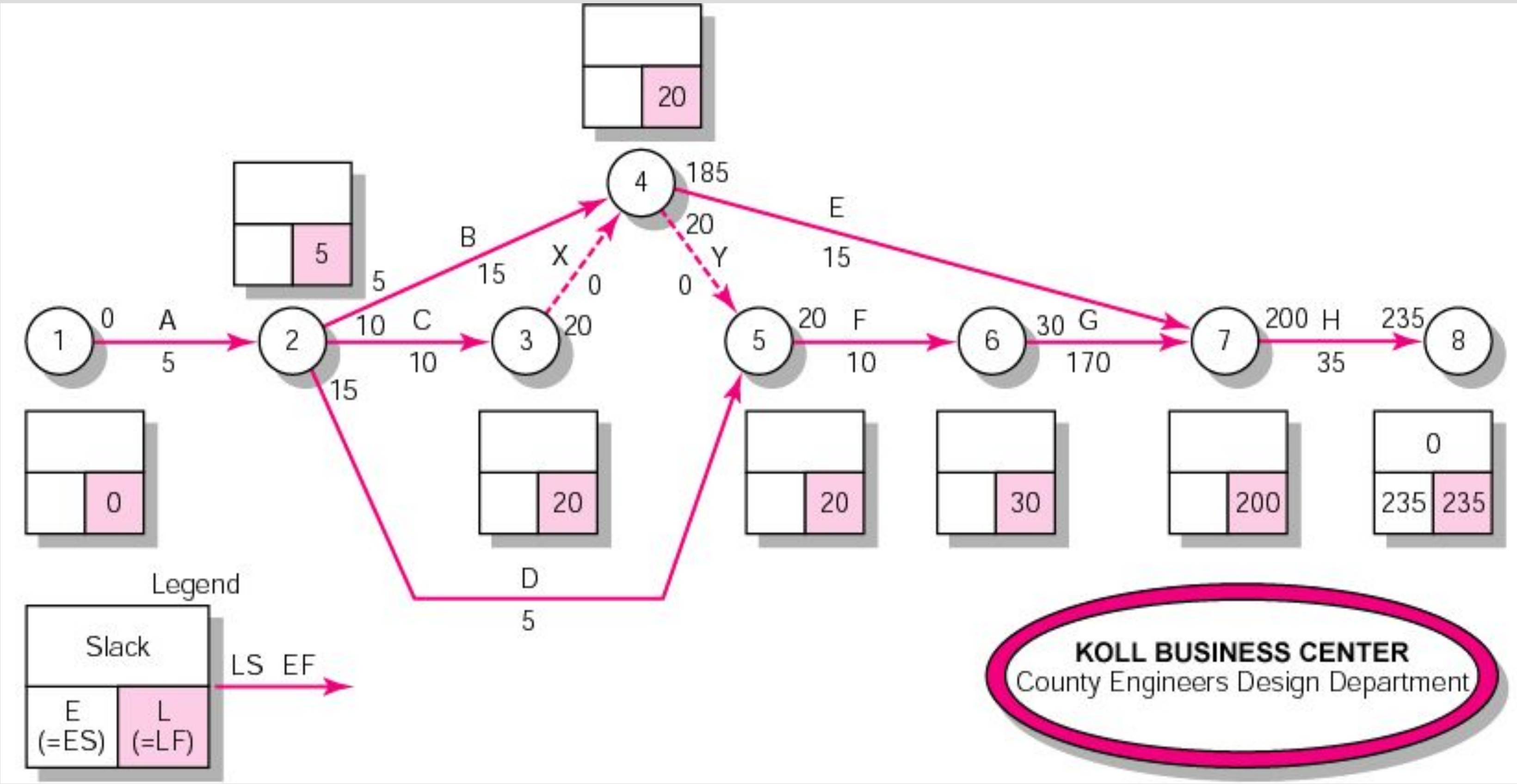
Activity-on-Arrow Network



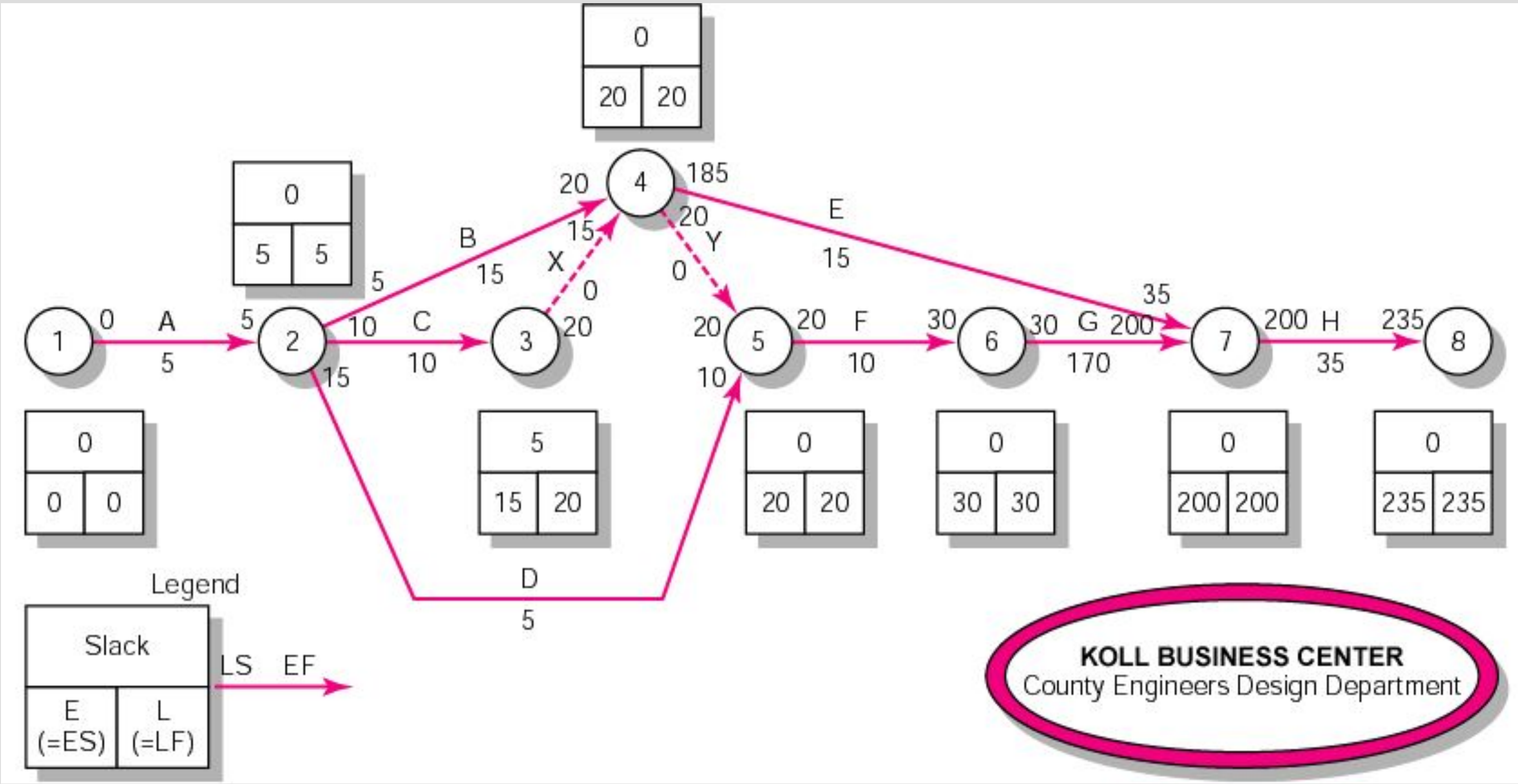
Activity-on-Arrow Network
Forward Pass



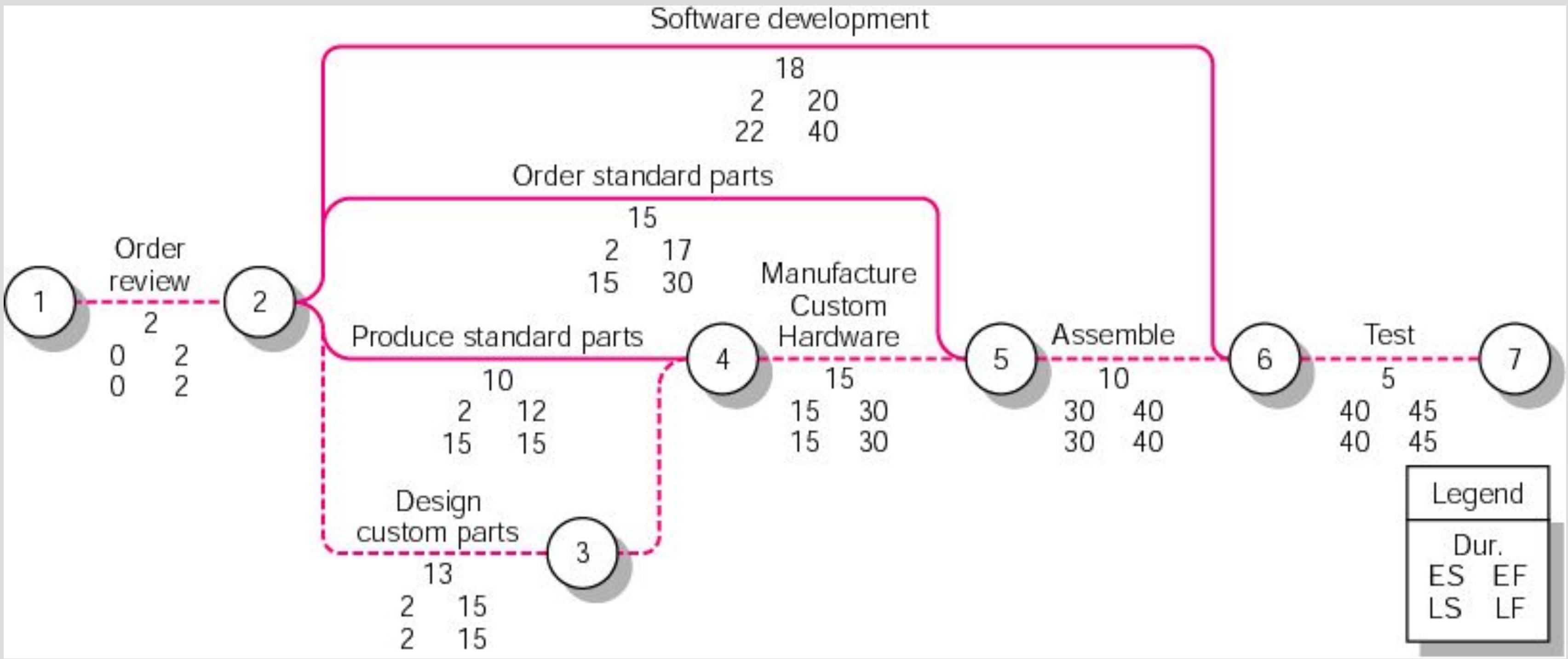
Activity-on-Arrow Network Backward Pass



Activity-on-Arrow Network Backward Pass, Forward Pass, and Slack



Air Control Inc. Custom Order Project—AOA Network Diagram



Comparison of AON and AOA Methods

AON Method

Advantages

1. No dummy activities are used.
2. Events are not used.
3. AON is easy to draw if dependencies are not intense.
4. Activity emphasis is easily understood by first-level managers.
5. The CPM approach uses deterministic times to construct networks.

Disadvantages

1. Path tracing by activity number is difficult. If the network is not available, computer outputs must list the predecessor and successor activities for each activity.
2. Network drawing and understanding are more difficult when dependencies are numerous.

AOA Method

Advantages

1. Path tracing is simplified by activity/event numbering scheme.
2. AOA is easier to draw if dependencies are intense.
3. Key events or milestones can easily be flagged.

Disadvantages

1. Use of dummy activities increases data requirements.
2. Emphasis on events can detract from activities. Activity delays cause events and projects to be late.

PERT

- PERT is based on the assumption that an activity's duration follows a probability distribution instead of being a single value
- Three time estimates are required to compute the parameters of an activity's duration distribution:
  - pessimistic time (t_p) - the time the activity would take if things did not go well
  - most likely time (t_m) - the consensus best estimate of the activity's duration
  - optimistic time (t_o) - the time the activity would take if things did go well

Mean (expected time): $$t_e = \frac{t_p + 4\,t_m + t_o}{6}$$

Variance: $$V_t = \sigma^2 = \left(\frac{t_p - t_o}{6}\right)^2$$

PERT analysis

- Draw the network.
- Analyze the paths through the network and find the critical path.
- The length of the critical path is the mean of the project duration probability distribution which is assumed to be normal
- The standard deviation of the project duration probability distribution is computed by adding the variances of the critical activities (all of the activities that make up the critical path) and taking the square root of that sum
- Probability computations can now be made using the normal distribution table.

Probability computation

Determine probability that project is completed within specified time

$$Z = \frac{x - \mu}{\sigma}$$

where $\mu = t_p$ = project mean time

σ = project standard mean time

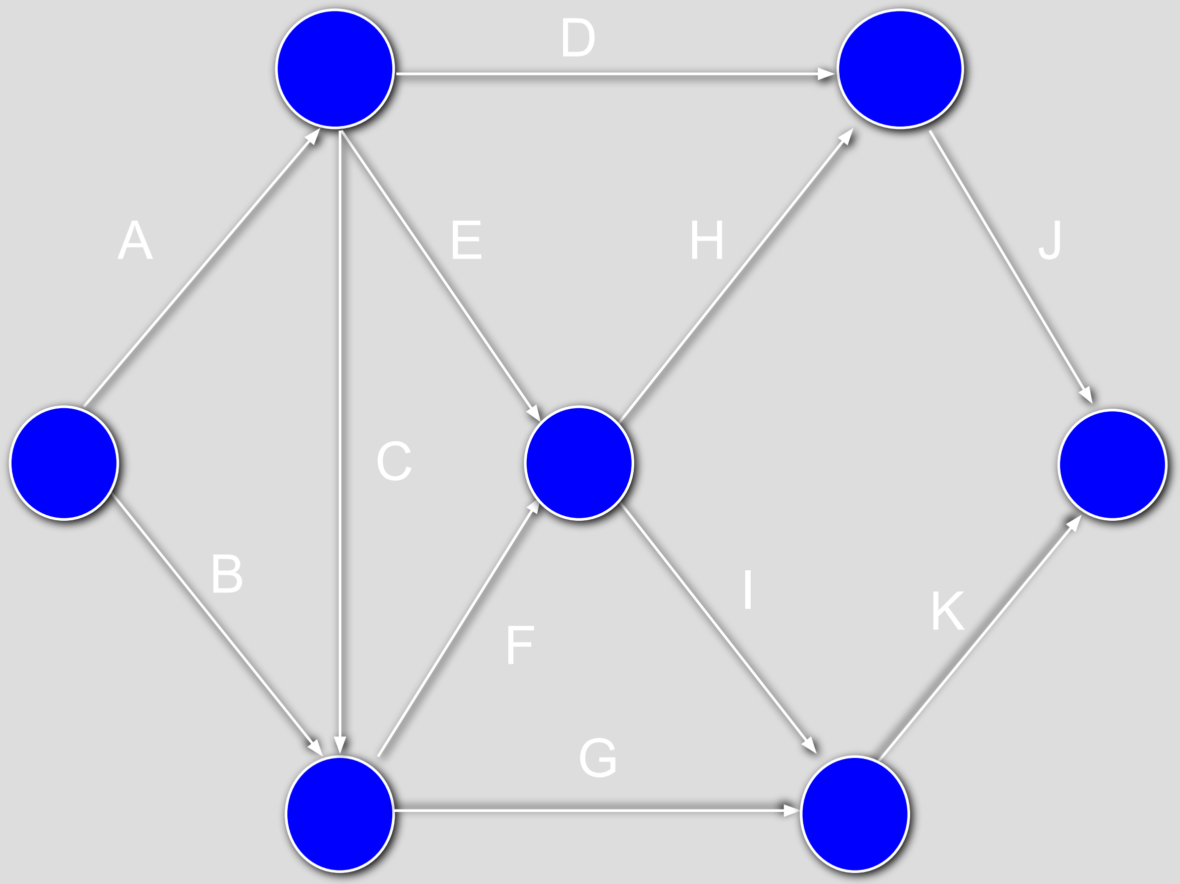
x = (proposed) specified time

PERT Example

Activity	Immed. Predec.	Optimistic Time (Hr.)	Most Likely Time (Hr.)	Pessimistic Time (Hr.)
A	--	4	6	8
B	--	1	4.5	5
C	A	3	3	3
D	A	4	5	6
E	A	0.5	1	1.5
F	B,C	3	4	5
G	B,C	1	1.5	5
H	E,F	5	6	7
I	E,F	2	5	8
J	D,H	2.5	2.75	4.5
K	G,I	3	5	7

PERT Example

PERT Network



PERT Example

Activity	Expected Time	Variance
A	6	4/9
B	4	4/9
C	3	0
D	5	1/9
E	1	1/36
F	4	1/9
G	2	4/9
H	6	1/9
I	5	1
J	3	1/9
K	5	4/9

PERT Example

Activity	ES	EF	LS	LF	Slack
A	0	6	0	6	0 *critical
B	0	4	5	9	5
C	6	9	6	9	0 *
D	6	11	15	20	9
E	6	7	12	13	6
F	9	13	9	13	0 *
G	9	11	16	18	7
H	13	19	14	20	1
I	13	18	13	18	0 *
J	19	22	20	23	1
K	18	23	18	23	0 *

PERT Example

$$V_{path} = V_A + V_C + V_F + V_I + V_K$$
$$= 4/9 + 0 + 1/9 + 1 + 4/9$$
$$= 2$$

$$\sigma_{path} = 1.414$$

$$z = (24 - 23)/\sigma = (24\text{-}23)/1.414 = .71$$

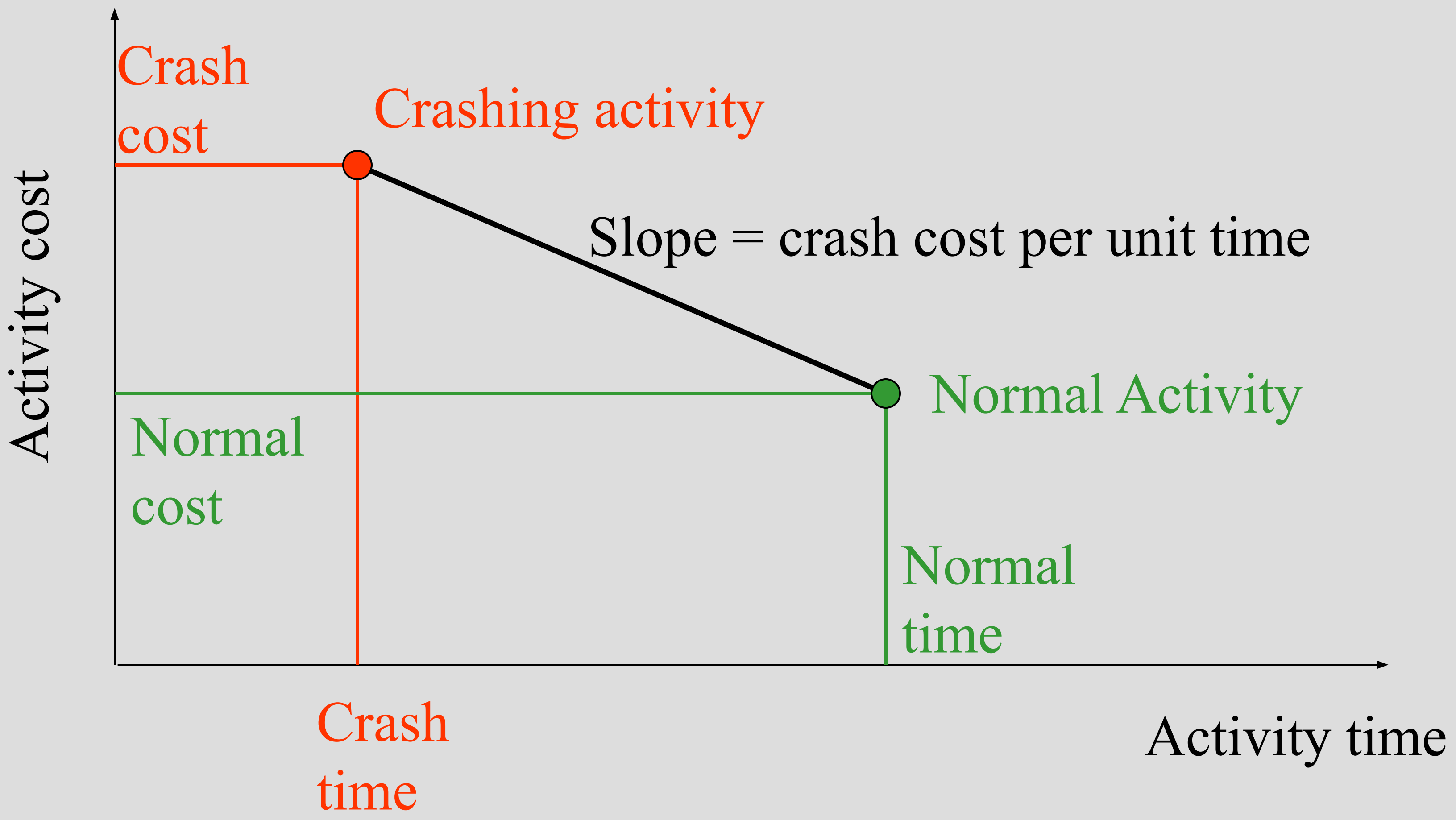
From the Standard Normal Distribution table:

$$P(z \leq .71) = .5 + .2612 = .7612$$

Project Crashing

- Crashing
  - reducing project time by expending additional resources
- Crash time
  - an amount of time an activity is reduced
- Crash cost
  - cost of reducing activity time
- Goal
  - reduce project duration at minimum cost

Activity crashing



Benefits of CPM/PERT

- Useful at many stages of project management
- Mathematically simple
- Give critical path and slack time
- Provide project documentation
- Useful in monitoring costs

CPM/PERT can answer the following important questions:

- How long will the entire project take to be completed? What are the risks involved?
- Which are the critical activities or tasks in the project which could delay the entire project if they were not completed on time?
- Is the project on schedule, behind schedule or ahead of schedule?
- If the project has to be finished earlier than planned, what is the best way to do this at the least cost?

Limitations to CPM/PERT

- Clearly defined, independent and stable activities
- Specified precedence relationships
- Over emphasis on critical paths
- Deterministic CPM model
- Activity time estimates are subjective and depend on judgment
- PERT assumes a beta distribution for these time estimates, but the actual distribution may be different
- PERT consistently underestimates the expected project completion time due to alternate paths becoming critical

To overcome the limitation, Monte Carlo simulations can be performed on the network to eliminate the optimistic bias

Practice Example

A social project manager is faced with a project with the following activities:

Activity Description	**Duration**
Social work team to live in village	5w
Social research team to do survey	12w
Analyse results of survey	5w
Establish mother & child health program	14w
Establish rural credit programme	15w
Carry out immunization of under fives	4w

Draw network diagram and show the critical path.
Calculate project duration.

Practice problem

Activity	Description	Duration
1-2	Social work team to live in village	5w
1-3	Social research team to do survey	12w
3-4	Analyse results of survey	5w
2-4	Establish mother & child health program	14w
3-5	Establish rural credit programme	15w
4-5	Carry out immunization of under fives	4w